

Summary of the 1st AIAA CFD Transition Modeling and Prediction Workshop

Jim Coder

University of Tennessee Knoxville

Advanced Modeling & Simulation (AMS) Seminar Series
NASA Ames Research Center, March 4, 2021

Workshop Objectives

- Assess the current state of the art in laminar-turbulent transition prediction in an industrial CFD environment
- Determine and document best practices for transitional flow simulations
- Verify transition/turbulence model implementations
- Encourage risk taking by participants and promote improvements to CFD prediction capabilities

Organizing Committee

- Jim Coder (University of Tennessee)
- Doug Stefanski (Oak Ridge National Laboratory)
- Jim Baeder (University of Maryland)
- Meelan Choudhari (NASA Langley)
- Marie Denison (NASA Ames)
- Matthew Tufts (AFRL)
- Andreas Krumbein (DLR)
- Greg Delattre (ONERA)

Agenda Recap

Day One – January 21, 2021	
<i>All times listed are Eastern Standard Time</i>	
10:00 – 10:05 AM	OPENING REMARKS
10:05 – 10:35 AM	CRM-NLF Overview Michelle Lynde (NASA Langley Research Center)
10:35 – 10:50 AM	Structured Mesh Families Jim Coder (University of Tennessee) and Marie Denison (NASA Ames)
10:50 – 11:05 AM	CRM-NLF Unstructured Mesh Family Steve Karman (Pointwise)
11:05 – 11:15 AM	BREAK
11:15 – 11:30 AM	Yuki Ide (JAXA)
11:30 – 11:45 AM	Norman Krimmelbein (DLR)
11:45 – 12:00 PM	Donato de Rosa (CIRA)
12:00 – 12:45 PM	BREAK
12:45 – 1:00 PM	Andrea Ciarella (Aircraft Research Association)
1:00 – 1:15 PM	Eric Laurendreau and Benjamin Barouillet (Polytechnique Montreal); Hong Yang and M. Charles-Armand Tatossian (Bombardier Aerospace)
1:15 – 1:30 PM	Balaji Venkatachari and Nathan Hildebrand (NASA Langley Research Center)
1:30 – 1:45 PM	Mike Piotrowski (NASA Ames); Dave Zingg (University of Toronto)
1:45 – 2:00 PM	Yong Su Jung, Bumseok Lee and Jim Baeder (University of Maryland)
2:00 – 2:15 PM	Marie Denison (NASA Ames Research Center)
2:15 – 2:45 PM	BREAK
2:45 – 3:15 PM	SPECIAL PRESENTATION Geza Schrauf (DLR)
3:15 – 4:00 PM	OPEN DISCUSSION

Day Two – January 22, 2021	
<i>All times listed are Eastern Standard Time</i>	
9:00 – 9:30 AM	PLENARY TALK Thomas Streit (DLR)
9:30 – 9:45 AM	Sang Hyun Park (KAIST)
9:45 – 10:00 AM	Gregory Delattre (ONERA)
10:00 – 10:15 AM	Zhixiang Xiao and Song Fu (Tsinghua University)
10:15 – 10:30 AM	Krishna Zore (ANSYS)
10:30 – 10:45 AM	BREAK
10:45 – 11:00 AM	Samet Cakmakcioglu (Turkish Aerospace); Unver Kaynak (Yildirim Beyazit University)
11:00 – 11:15 AM	Jan-Soren Fischer (Netherlands Aerospace Centre)
11:15 – 11:30 AM	Jiri Furst (CVUT)
11:30 – 11:45 AM	Rui Lopes and Luis Eça (IST); Maarten Kerkvliet and Serge Toxopeus (MARIN)
11:45 – 12:00 PM	Marian Zastawny and Chris Nelson (Siemens)
12:00 – 12:45 PM	BREAK
12:45 – 1:45 PM	EXPERIMENTALIST PANEL DISCUSSION Moderator: Dr. Richard Wlezien (Missouri S&T) Panelists: Marco Costantini (DLR); Austin Overmeyer (US Army CCDC AvMC); and Ed White (Texas A&M)
1:45 – 2:00 PM	BREAK
2:00 – 3:00 PM	DATA SUMMARY AND OPEN DISCUSSION

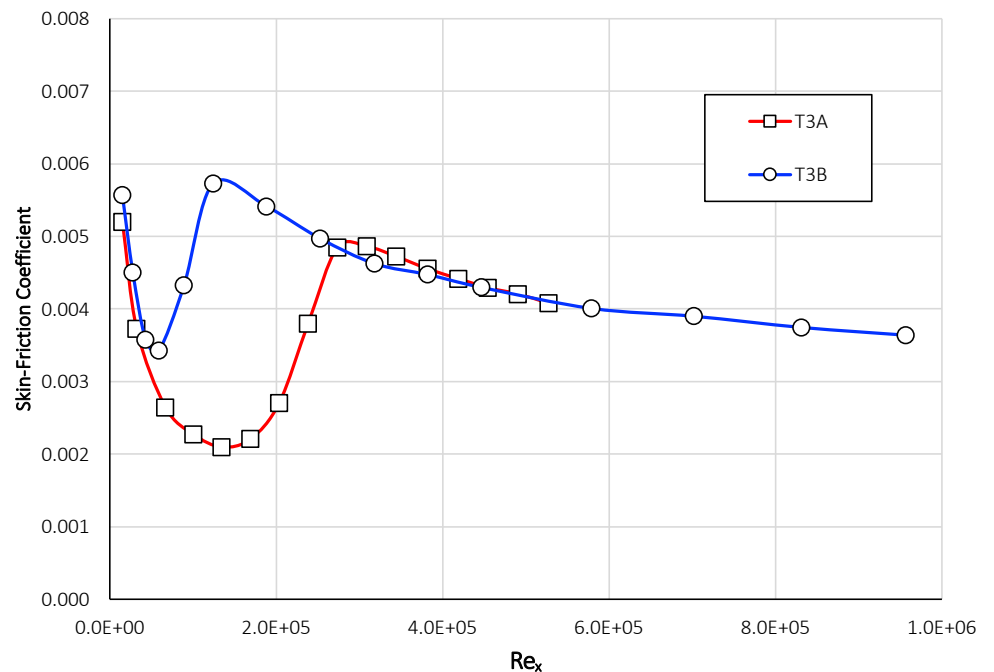
Case 1 – ZPG Flat Plate

Bypass transition cases based on ERCOFTAC T3A and T3B cases

Free-stream turbulence decay is a significant consideration

- Grid dimensions and inlet conditions tightly coupled
- Very large eddy-viscosity values required

Grid resolution studies requested for both cases

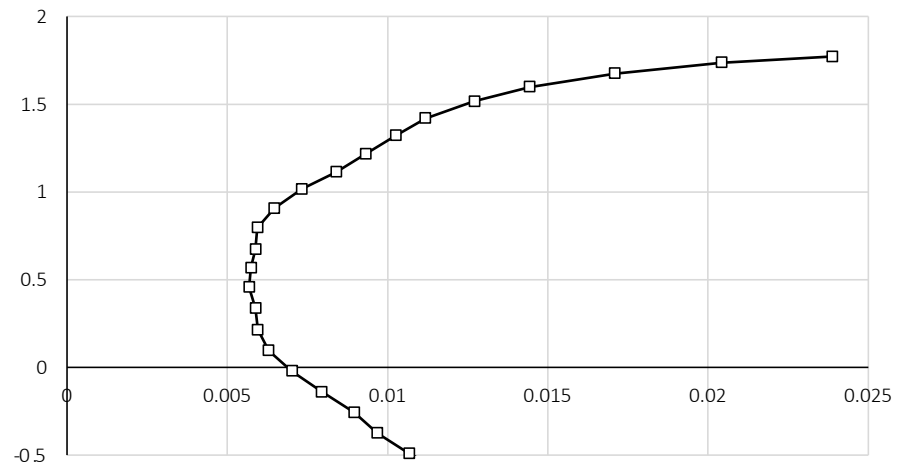
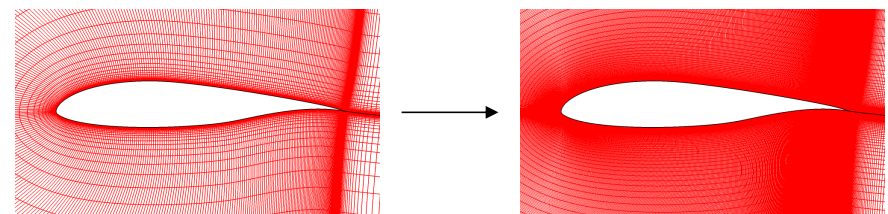


Case 2 – NLF(1)-0416 Airfoil

Two-dimensional case with natural transition under pressure gradient

- Force/moment and transition location data available from NASA LTPT experiment

Grid resolution study at two alphas and alpha sweep with medium-resolution grid requested



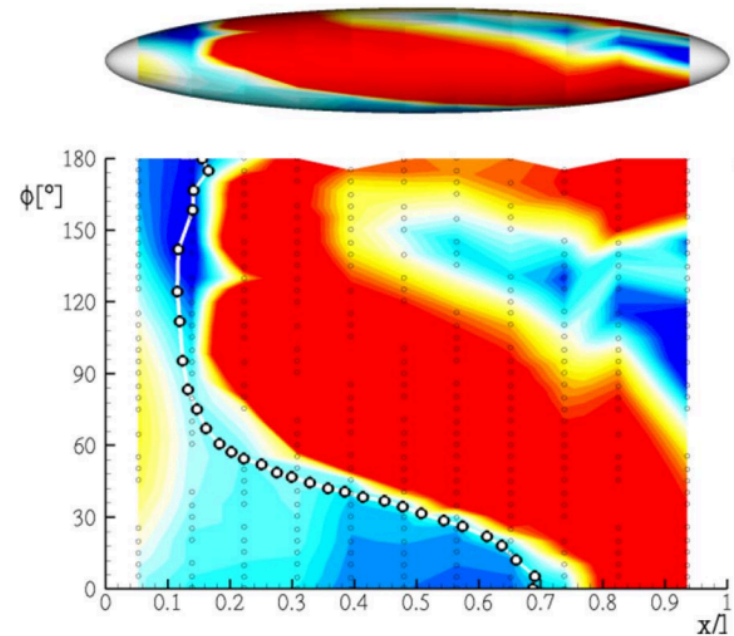
Case 3 – 6:1 Prolate Spheroid

Three-dimensional, axisymmetric geometry at inclination

- Experimental skin-friction measurements available

Both crossflow and streamwise mechanisms are prominent

Alpha sweep with medium-resolution grid requested with optional grid-refinement study



Case 4 – CRM-NLF

Continues the legacy and traceability of using the CRM for prediction workshops

Commercial transport wing designed to achieve extensive laminar flow at flight Reynolds numbers and lift coefficients

Three-dimensional, shock-dominated flow with multiple transition mechanisms

Grid-resolution study and alpha sweep requested



Data Summary Overview

- Data submitted by 18 participant teams ahead of workshop
 - 7 government, 4 industry, 7 academia
 - Additional representation from mixed teams
- Strong international participation
 - USA, Canada, Germany, UK, France, Portugal, Netherlands, Czechia, Italy, Turkey, India, China, Japan
 - Only 1 US university

Data Plotting

- Letters used as plotting symbols to denote specific participants in a random, but consistent, order
 - Participant key currently being withheld
- Data are color-coded to characterize the type of turbulence/transition model used
- **Each participant used a different model**
 - Prevented true code-to-code verification

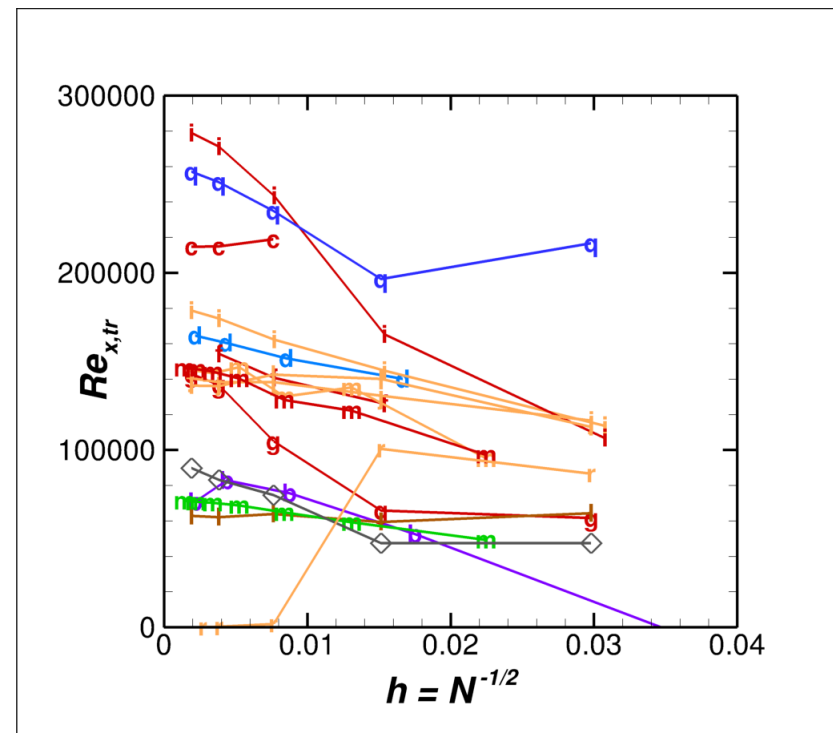
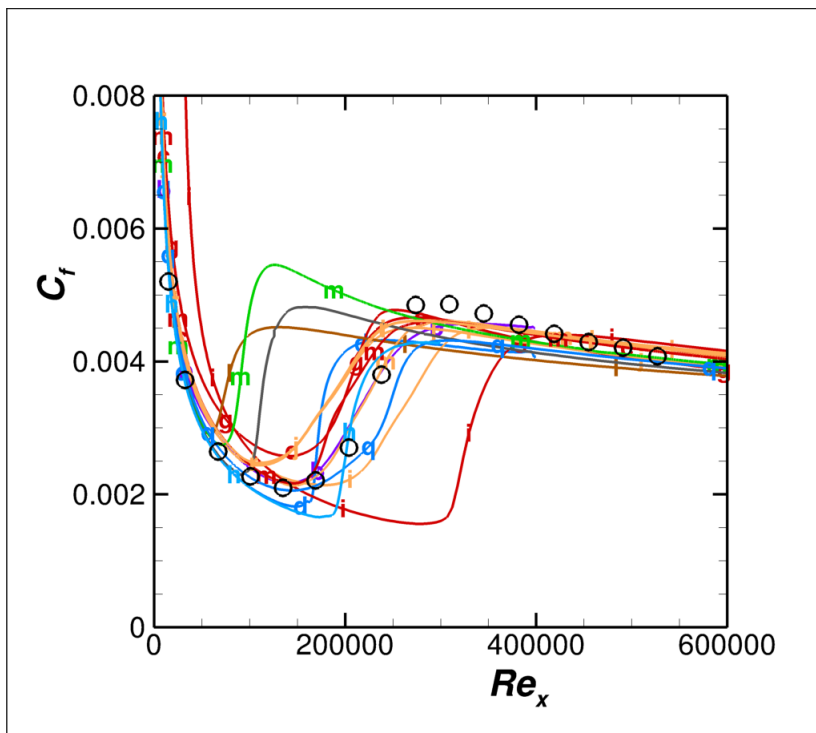
Turbulence and Transition Models

- Linear Stability with SA
- Linear Stability with SST and RSM
- γ - $\text{Re}_{\theta t}$ (LM2009) with SST
- γ (MS2015) with SST
- γ - $\text{Re}_{\theta t}$ (LM2009/LM2015)+CF(He) with SST and RSM
- γ - $\text{Re}_{\theta t}$ (LM2009)+CF+ with SST
- γ (MS2015)+CF(He) with SST
- AFT2017b with SA
- γ - $\text{Re}_{\theta t}$ (sLM2015) with SA
- MB + CF(He) with SA (Medida/Baeder)
- B-C with SA (Bas/Çakmakçioğlu: algebraic transition model)
- Lag Elliptic Model

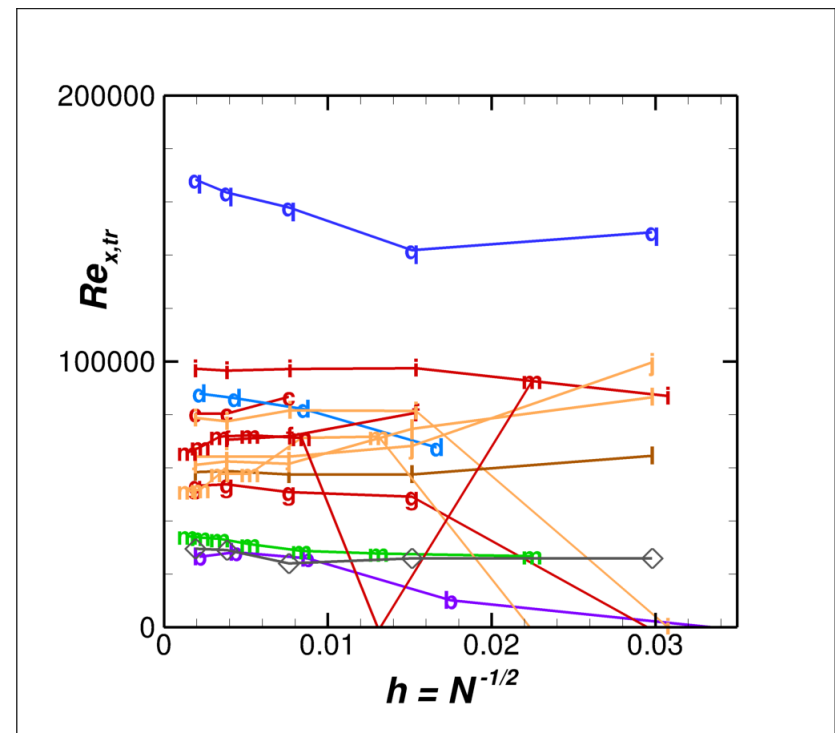
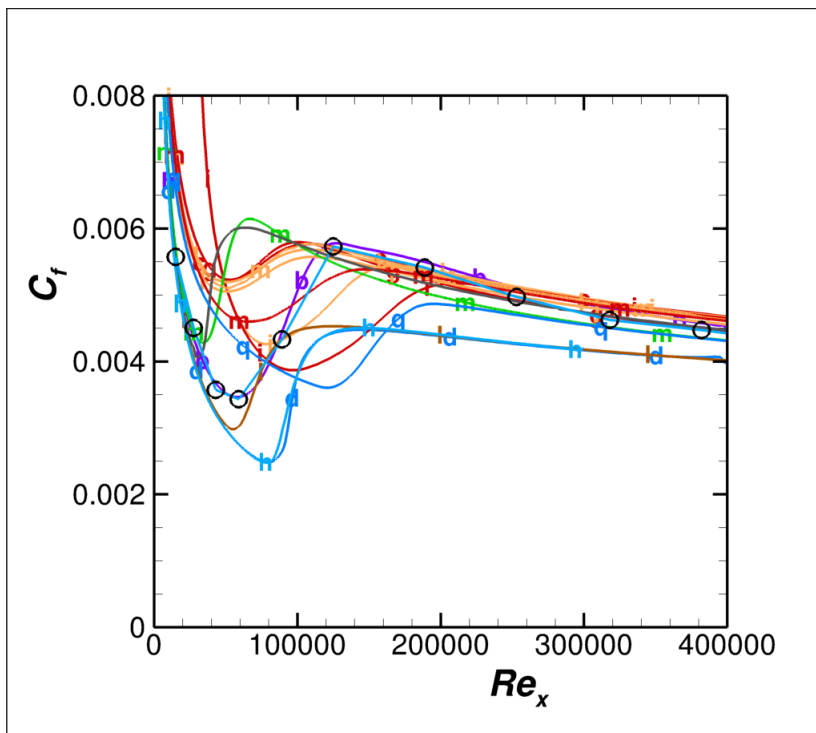
Case 1 – ZPG Flat Plate

- **Participant Data**
 - **13 data submittals**
 - 6 N. America, 5 Europe, 2 Asia
 - 5 Government, 4 Academia, 2 Industry, 2 Industry/Academia
 - **Grid types**
 - 10 committee-provided structured
 - 2 custom structured
 - 1 custom unstructured

Case 1A – T3A



Case 1B – T3B



Case 2 – NLF(1)-0416

- **Participant Data**

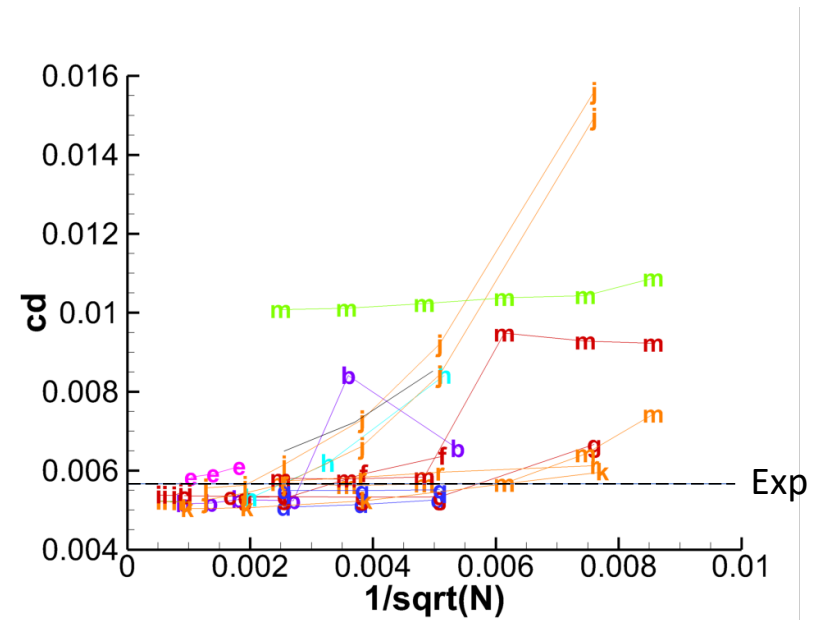
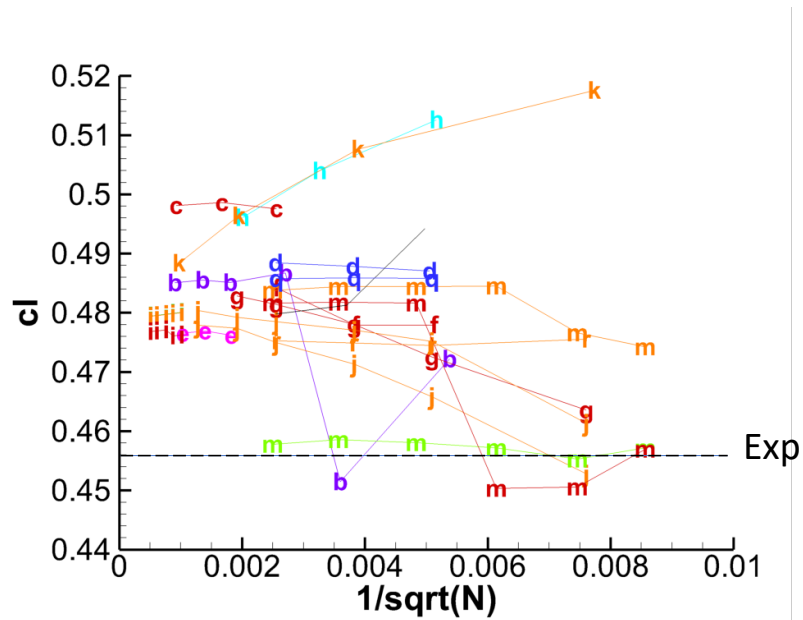
- **14 data submittals**

- 6 N. America, 6 Europe, 2 Asia
 - 6 Government, 4 Academia, 2 Industry, 2 Industry/Academia

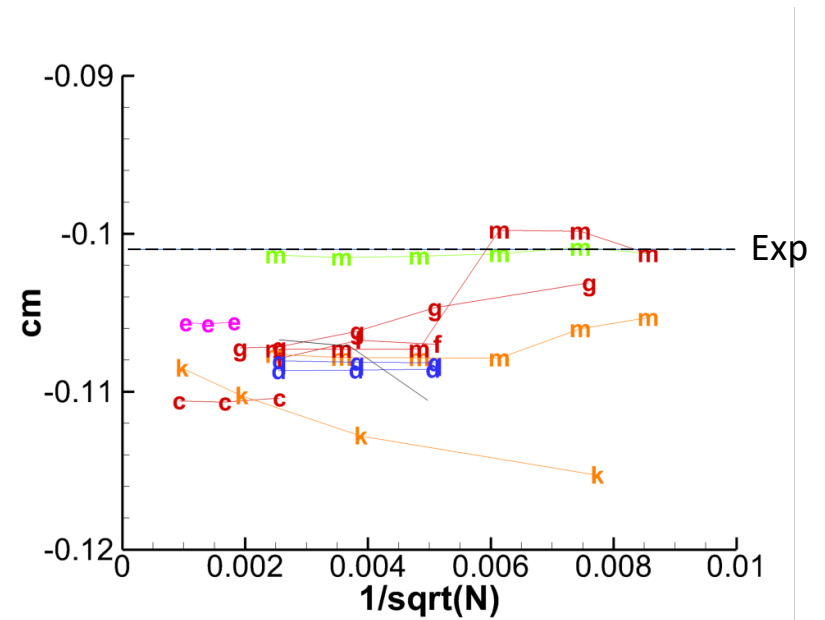
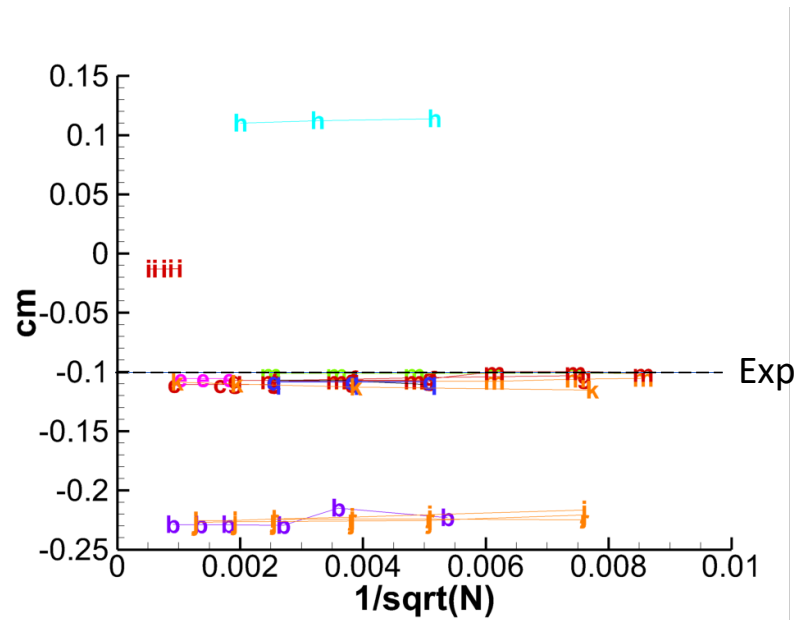
- **Grid types**

- 8 committee-provided structured
 - 4 custom structured
 - 2 custom unstructured

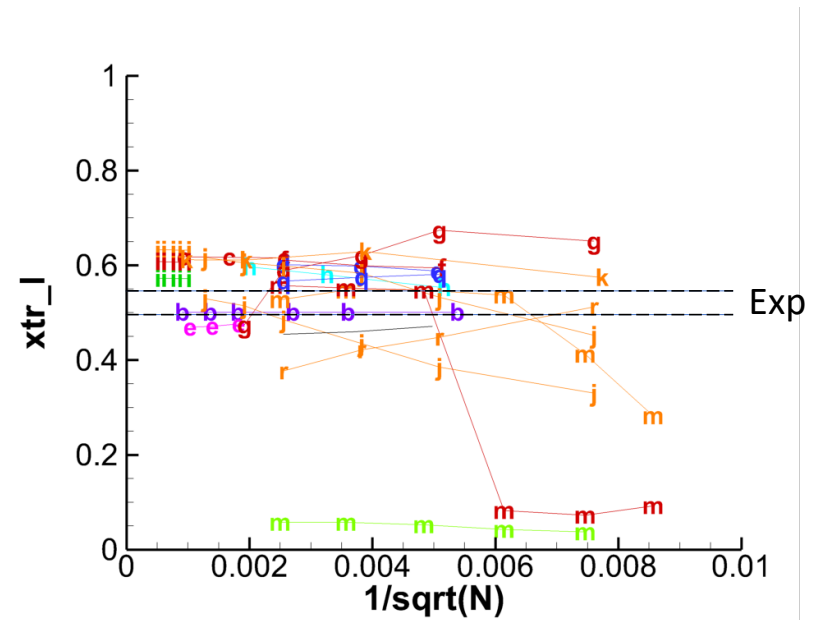
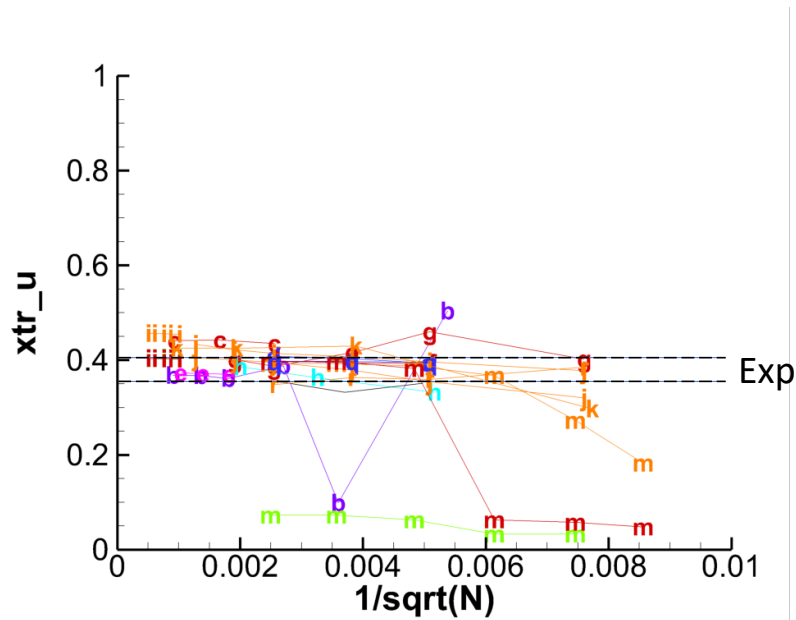
Case 2A – Grid Convergence ($\alpha = 0^\circ$)



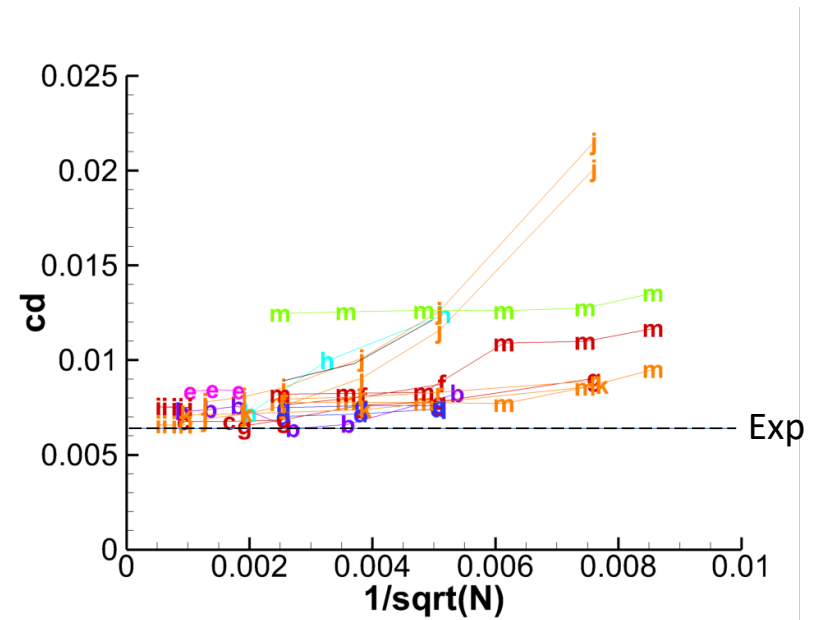
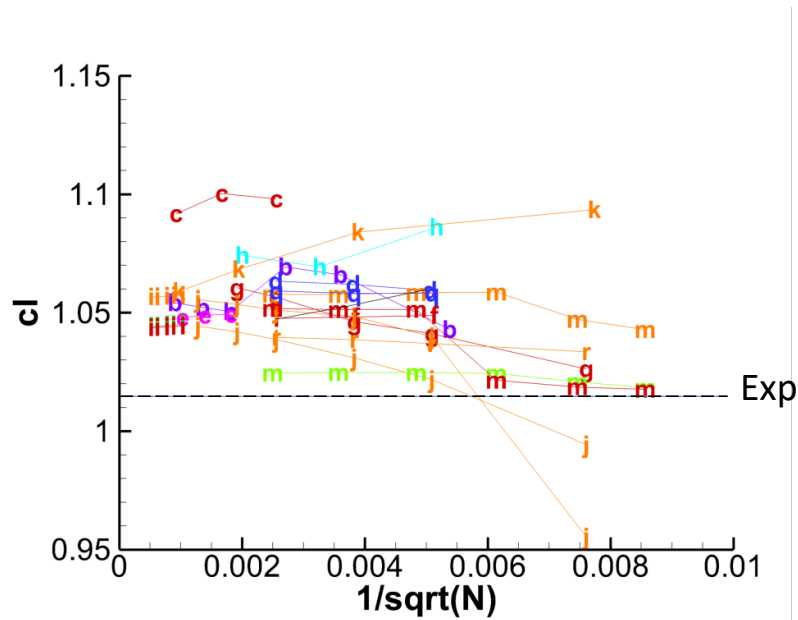
Case 2A – Grid Convergence ($\alpha = 0^\circ$)



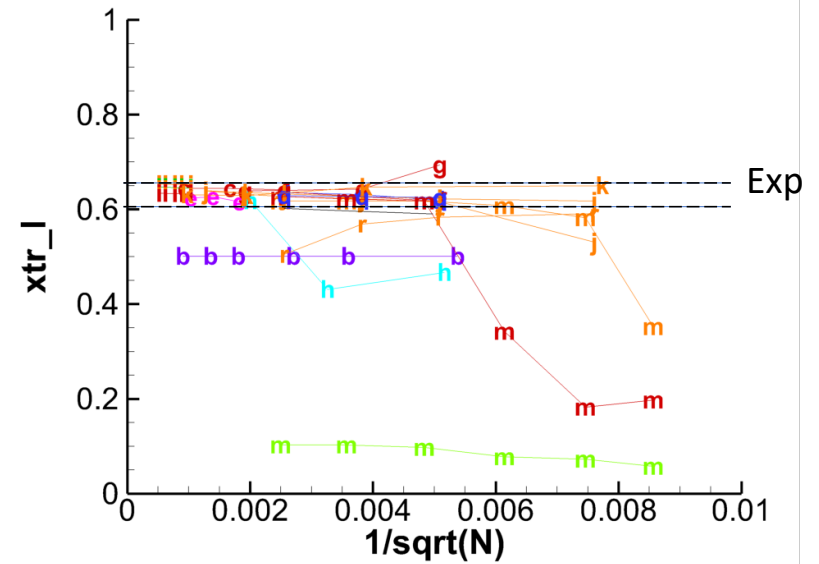
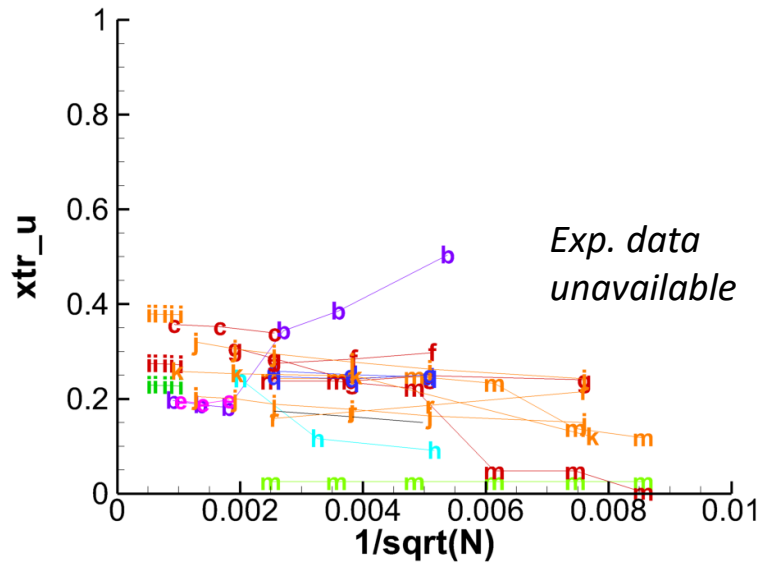
Case 2A – Grid Convergence ($\alpha = 0^\circ$)



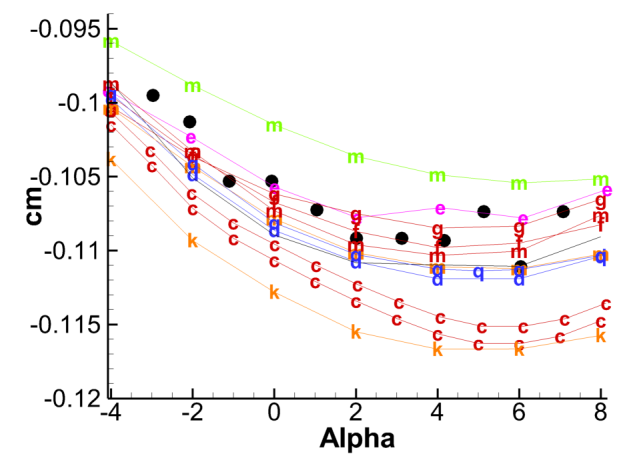
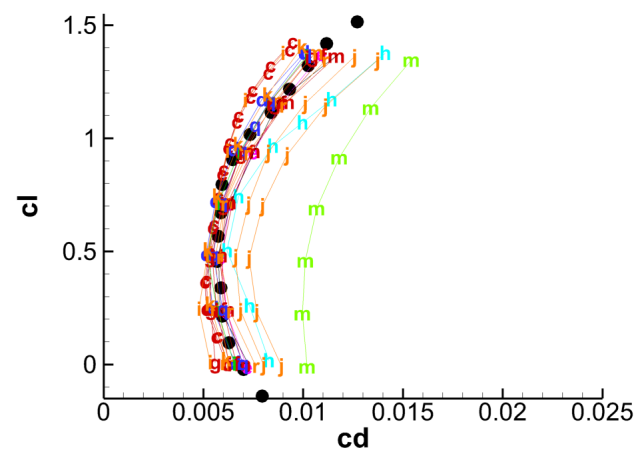
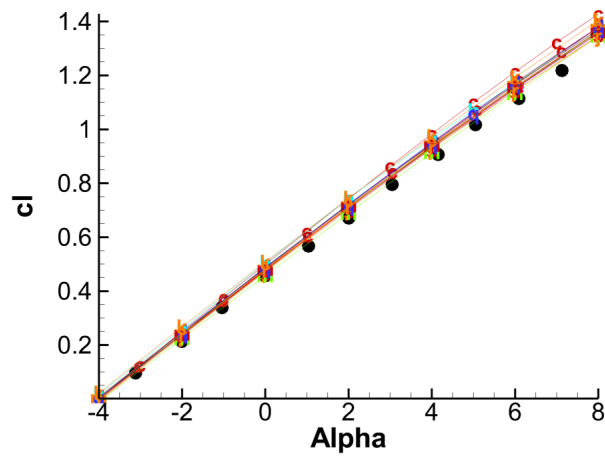
Case 2A – Grid Convergence ($\alpha = 5^\circ$)



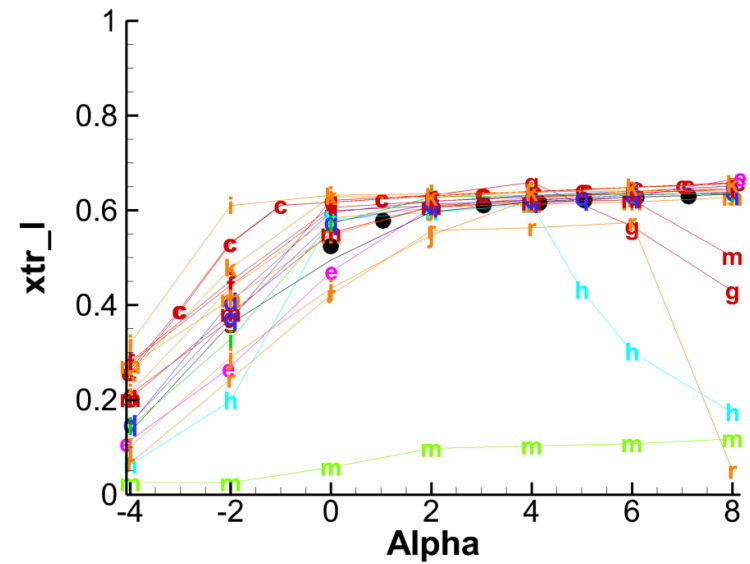
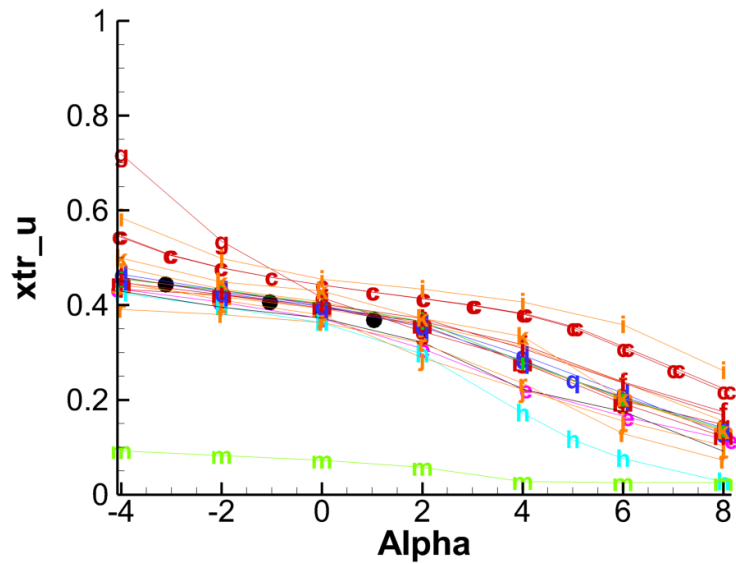
Case 2A – Grid Convergence ($\alpha = 5^\circ$)



Case 2B – Alpha Sweep



Case 2B – Alpha Sweep



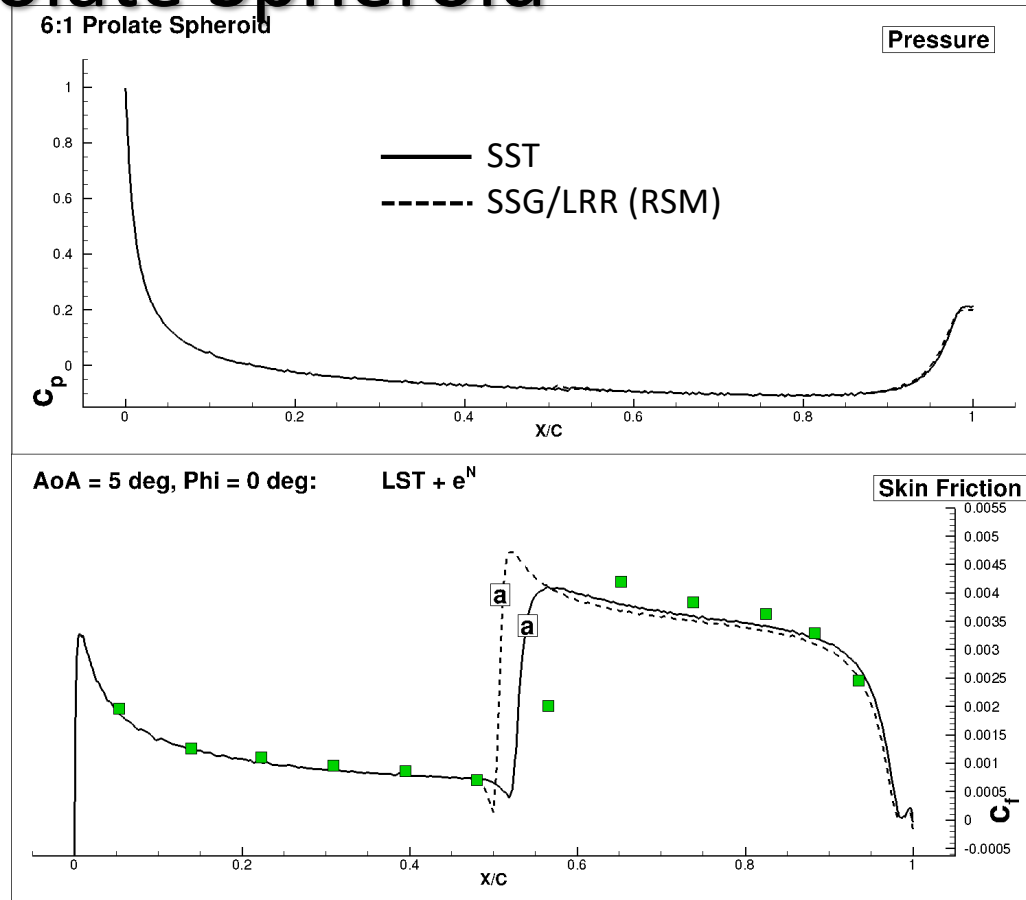
Case 3 – 6:1 Prolate Spheroid

- **Participant Data**
 - **11 data submittals**
 - 5 N. America, 3 Europe, 3 Asia
 - 3 Government, 5 Academia, 1 Industry, 2 Industry/Academia
 - **Grid types**
 - 3 committee-provided structured
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 - 3 custom structured
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Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 0 deg:

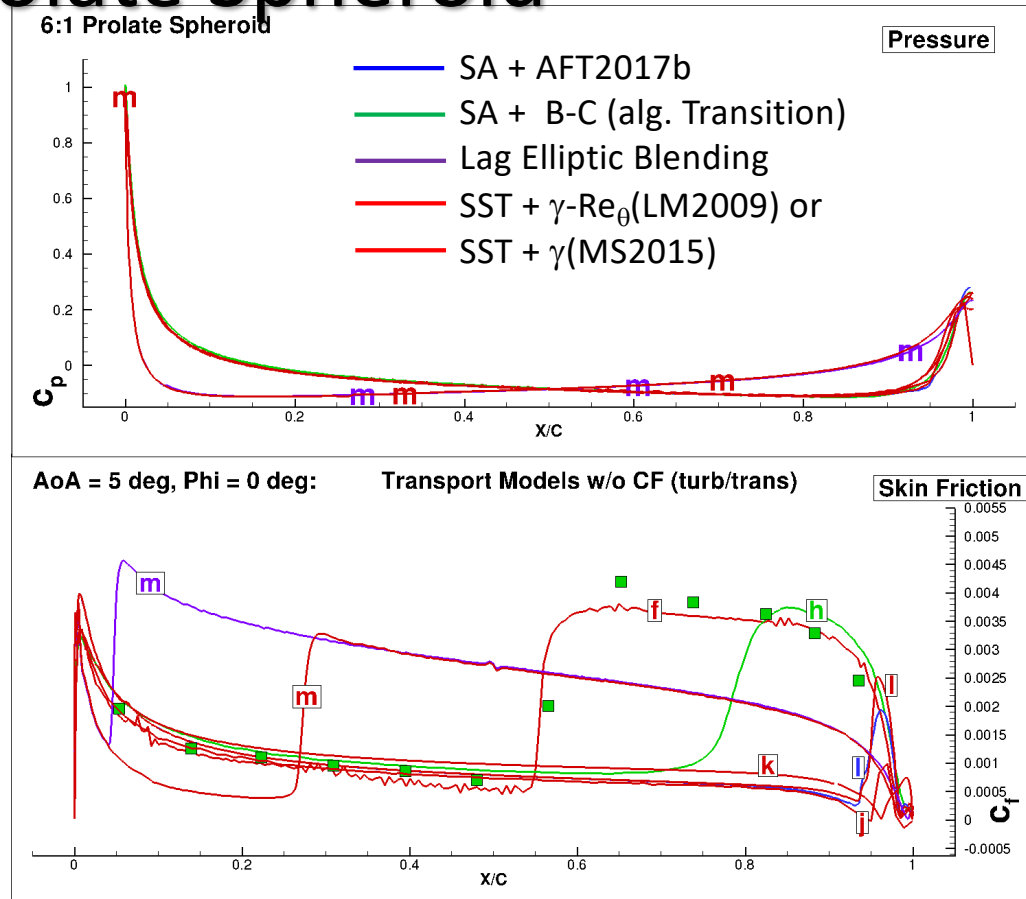
- LST + e^N



Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 0 deg:

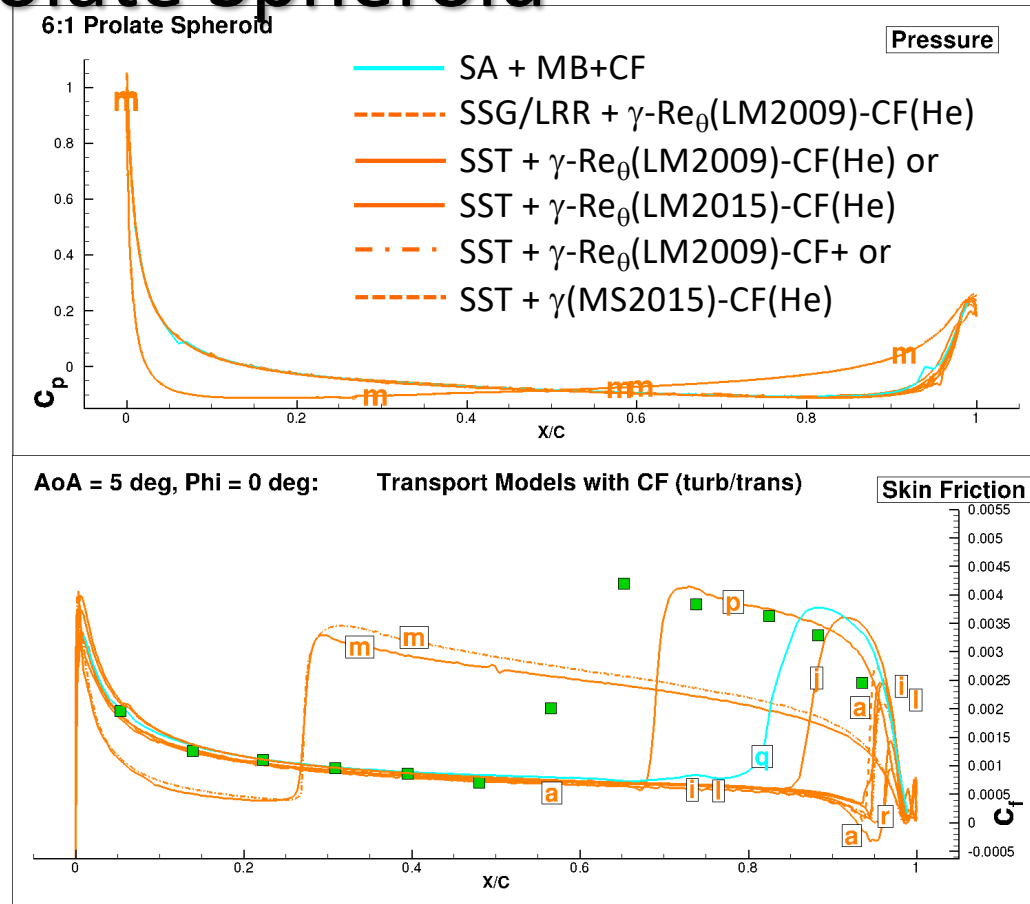
- Transport Models without CF Transition



Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 0 deg:

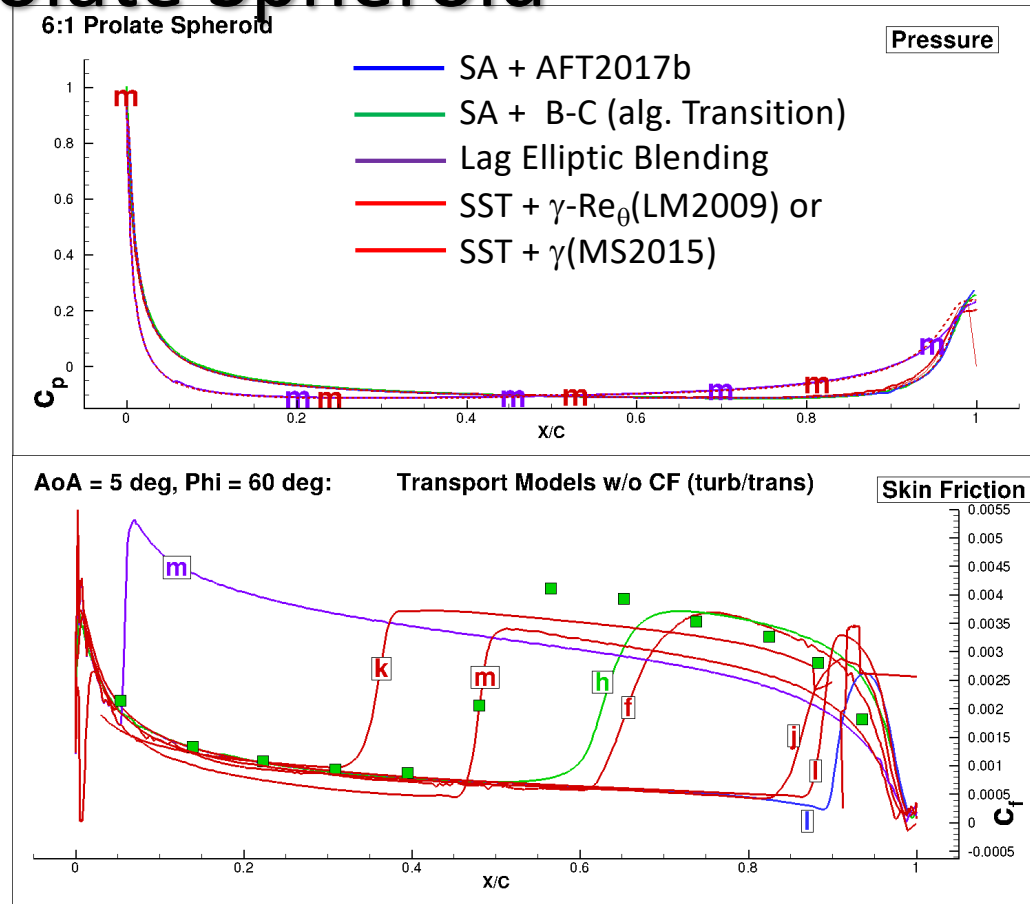
- Transport Models with
CF Transition



Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 60 deg:

- Transport Models without CF Transition

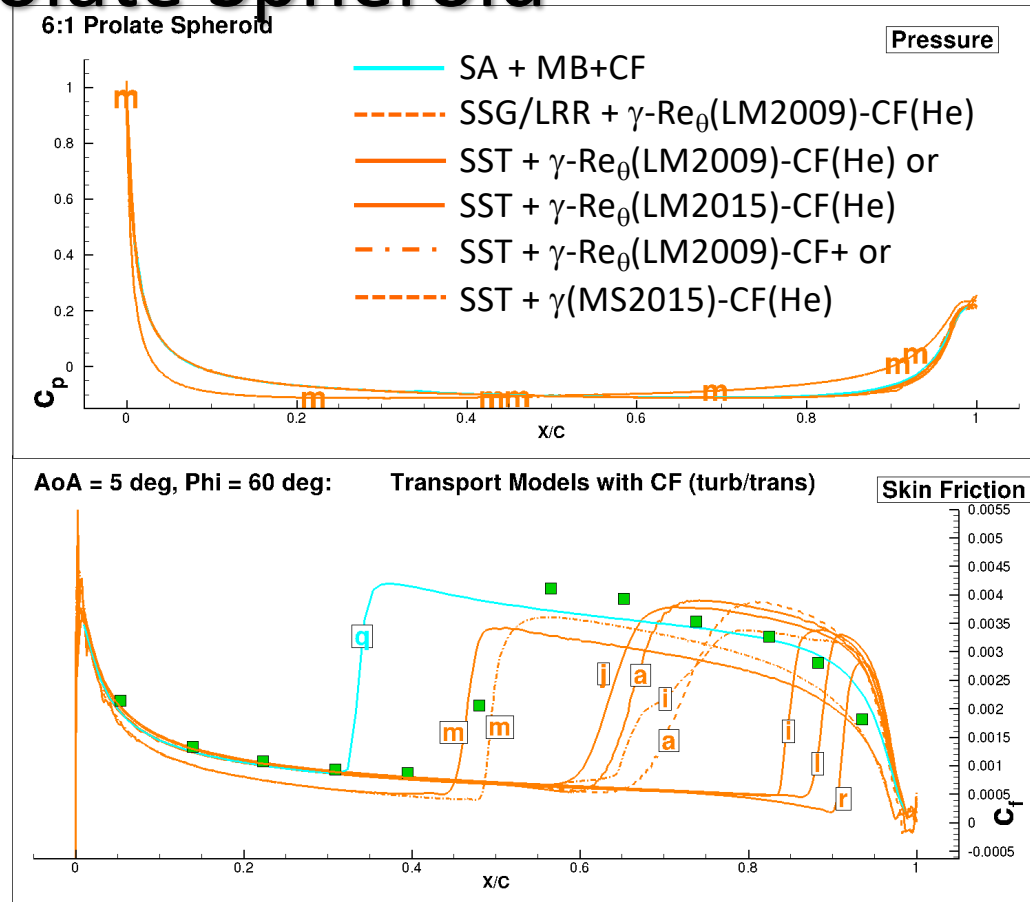


Error: „k“ → SST-2003 +
LM2009.mod + CF(HE).mod
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Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 60 deg:

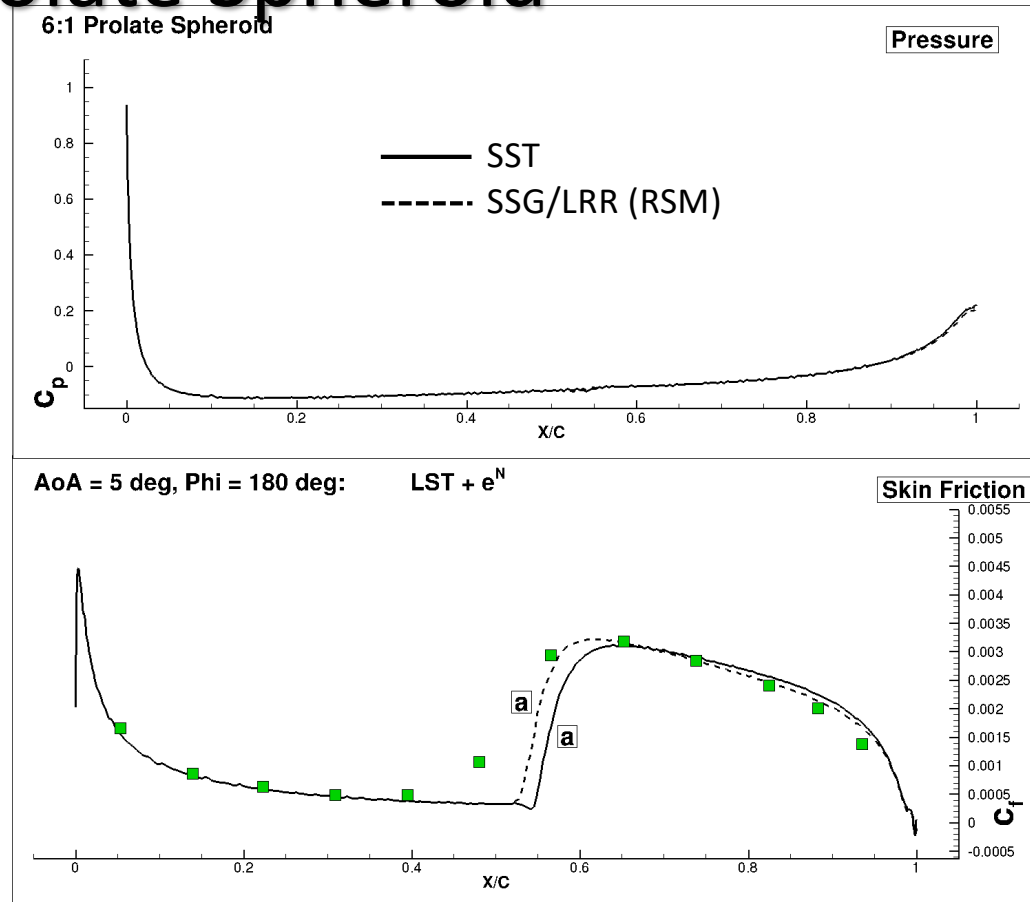
- Transport Models with
CF Transition



Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 180 deg:

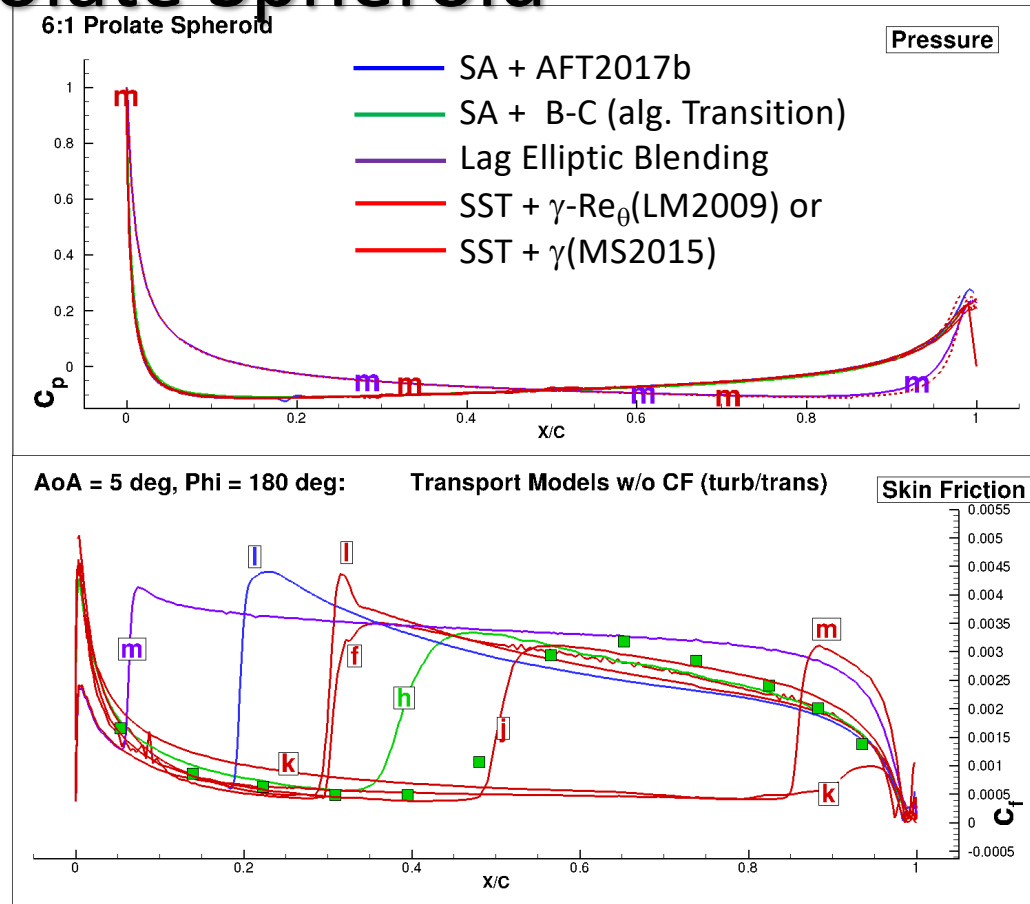
- LST + e^N



Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 180 deg:

- Transport Models without CF Transition

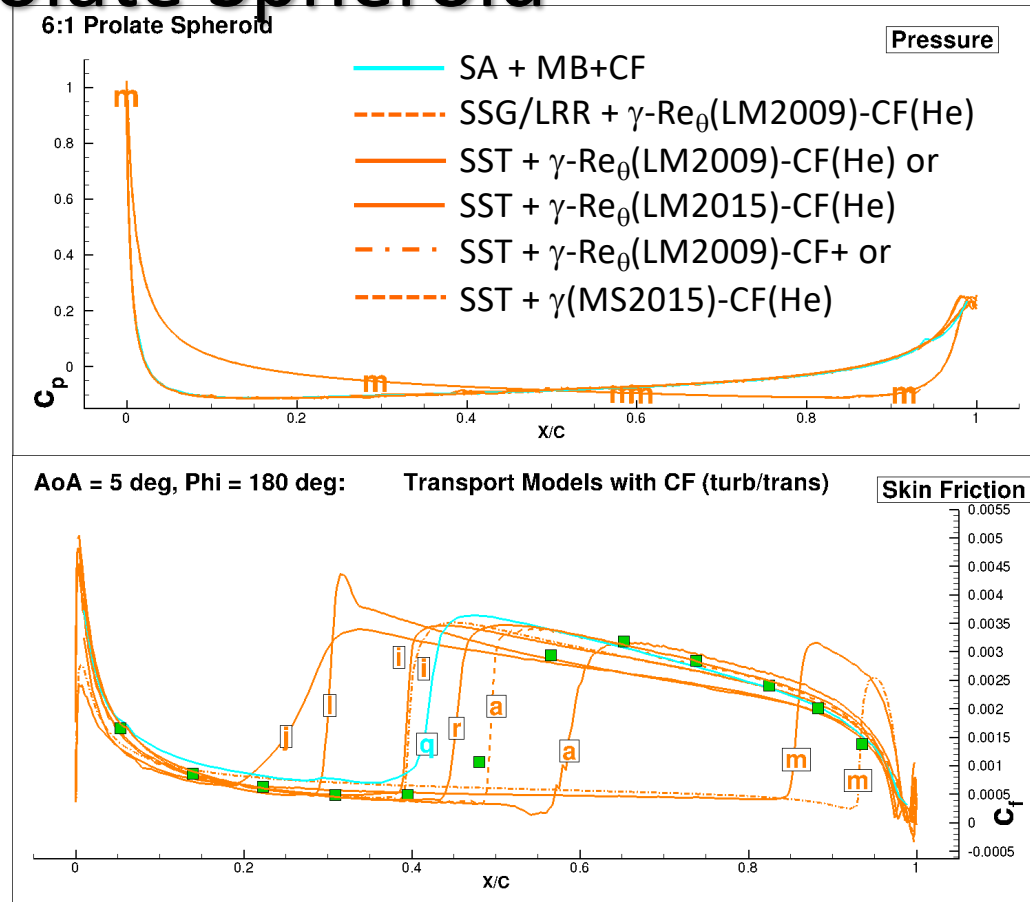


Error: „k“ → SST-2003 +
LM2009.mod + CF(HE).mod
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Case 3 – 6:1 Prolate Spheroid

AoA = 5 deg,
Phi = 180 deg:

- Transport Models with
CF Transition



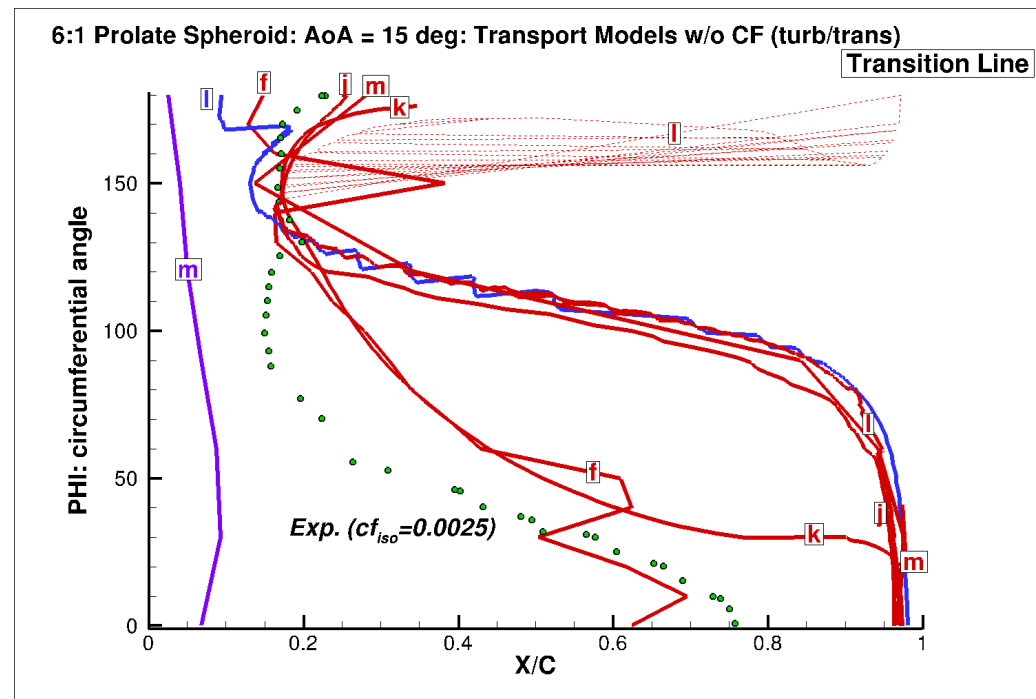
Case 3 – 6:1 Prolate Spheroid

AoA = 15 deg:

- Transport Models without CF Transition

- SA + AFT2017b
- SA + B-C (alg. Transition)
- Lag Elliptic Blending
- SST + γ - Re_θ (LM2009) or
- SST + γ (MS2015)

Error: „k“ → SST-2003 + LM2009.mod + CF(HE).mod
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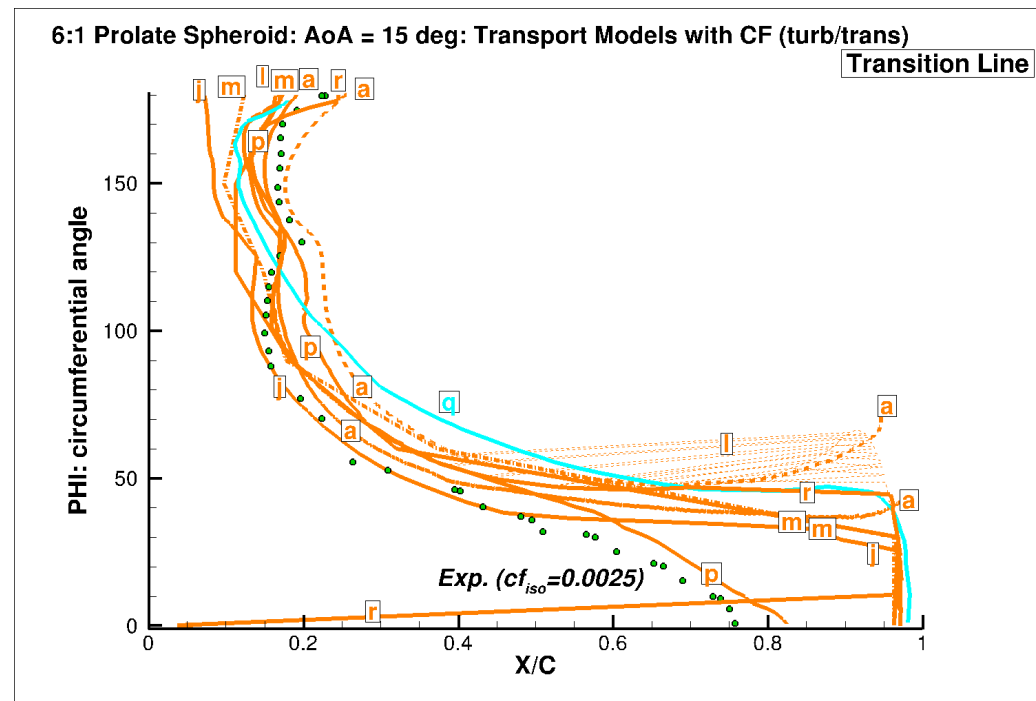


Case 3 – 6:1 Prolate Spheroid

AoA = 15 deg:

- Transport Models with CF Transition

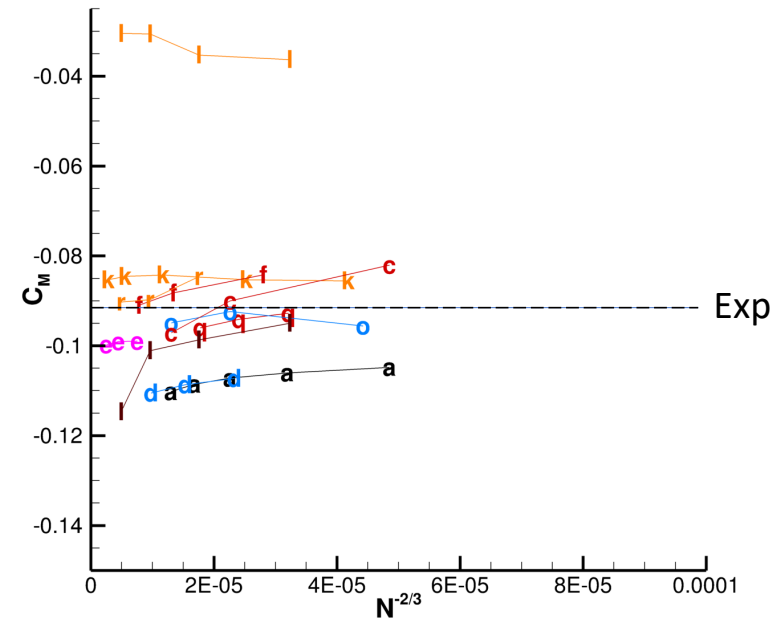
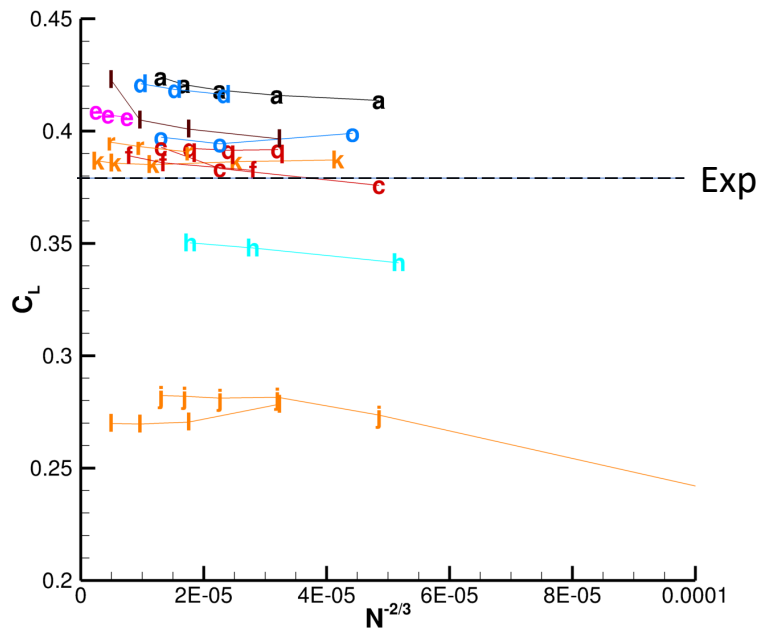
- SA + MB+CF
- - - SSG/LRR + γ -Re $_{\theta}$ (LM2009)-CF(He)
- SST + γ -Re $_{\theta}$ (LM2009)-CF(He) or
- SST + γ -Re $_{\theta}$ (LM2015)-CF(He)
- . - . SST + γ -Re $_{\theta}$ (LM2009)-CF+ or
- - - SST + γ (MS2015)-CF(He)



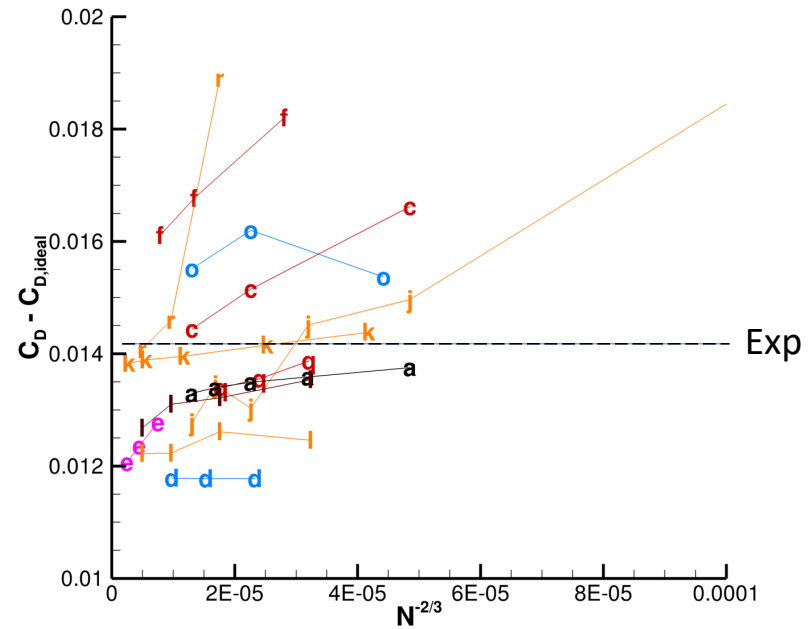
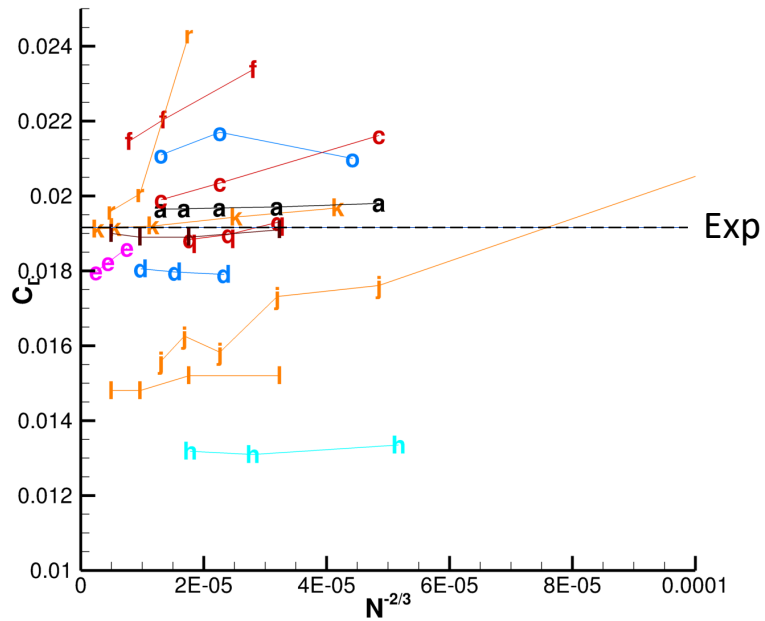
Case 4 – CRM-NLF

- **Participant Data**
 - **13 data submittals**
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 - **Grid types**
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Case 4A – Grid Convergence

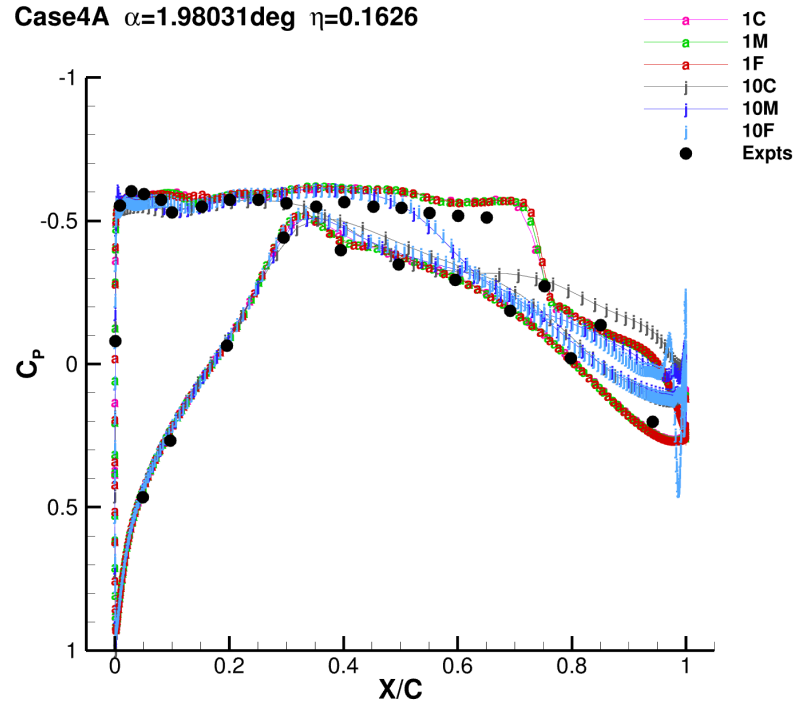


Case 4A – Grid Convergence

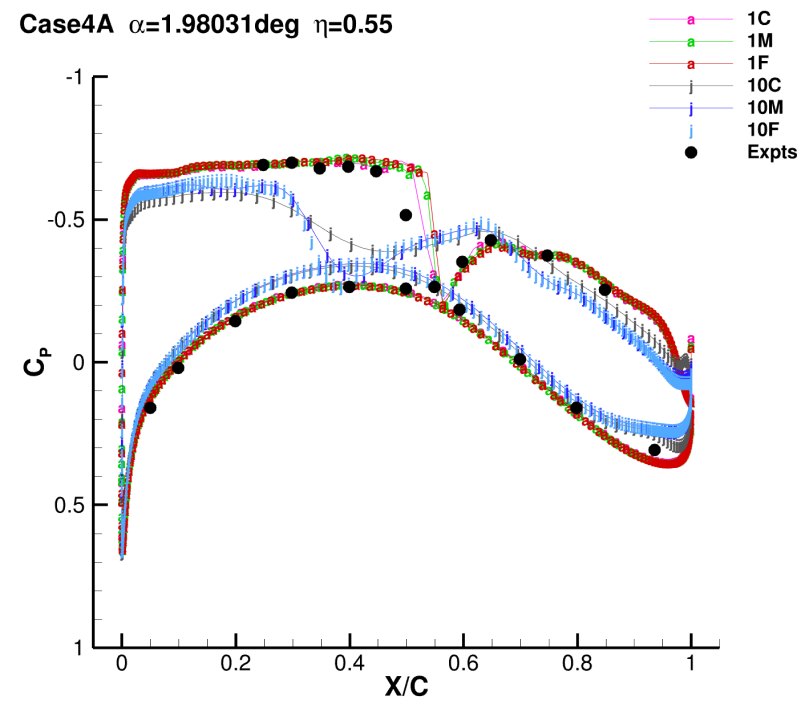


Case 4A – Grid Convergence

Case4A $\alpha=1.98031\text{deg}$ $\eta=0.1626$

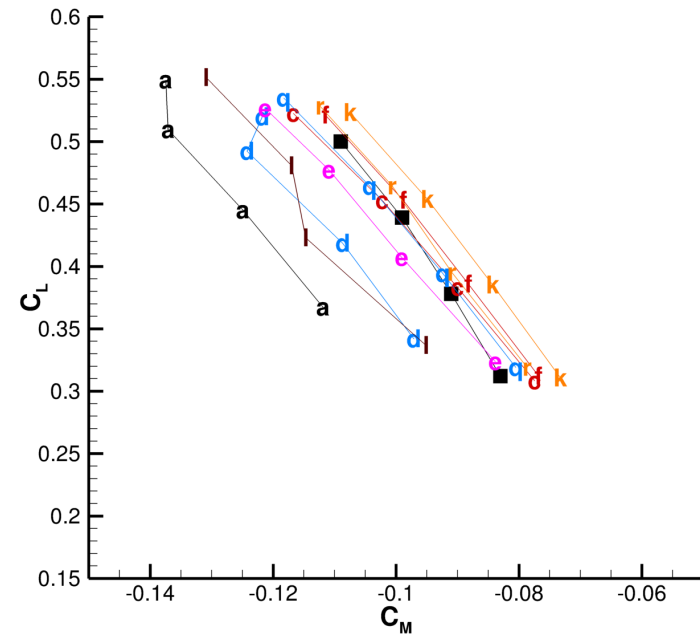
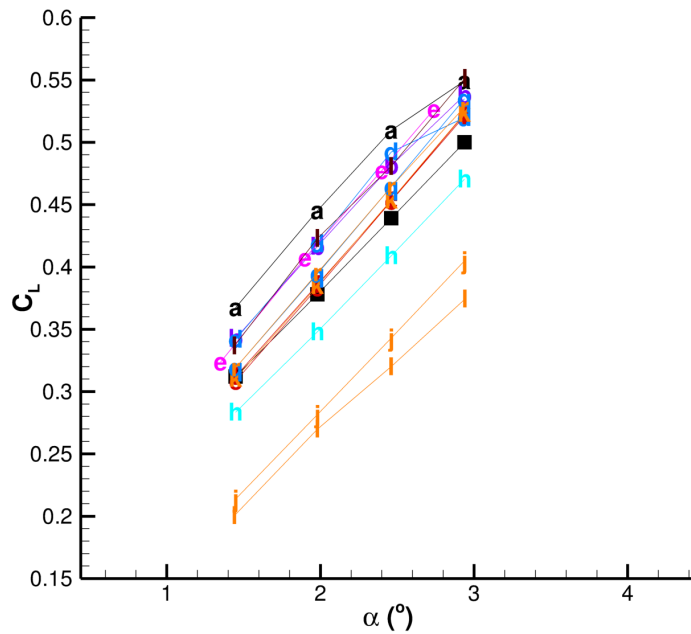


Case4A $\alpha=1.98031\text{deg}$ $\eta=0.55$

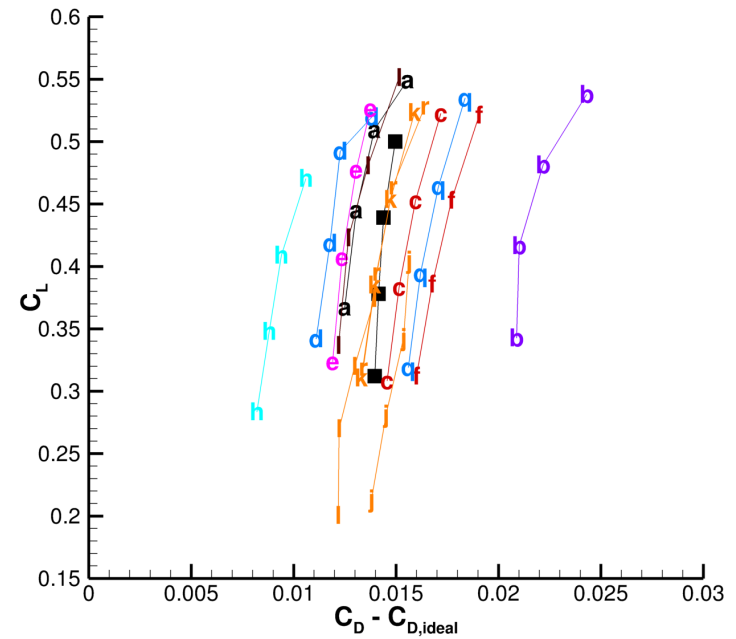
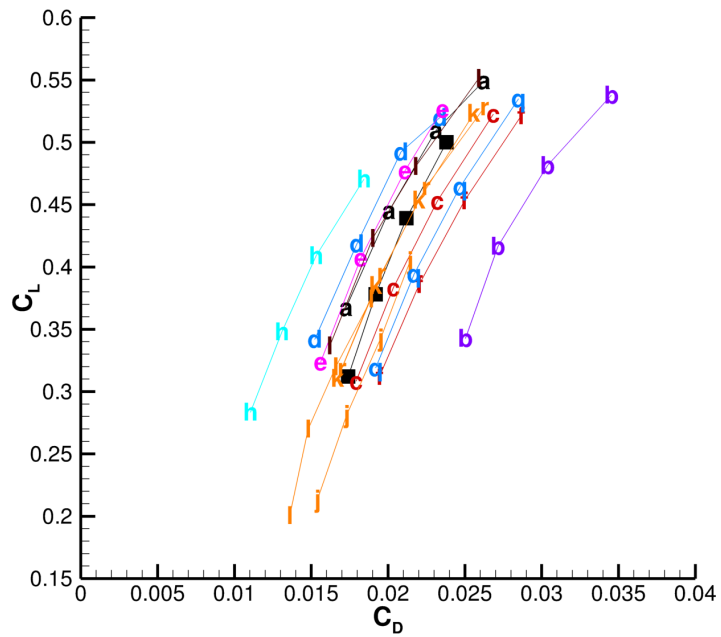


Case 4 – CRM-NLF

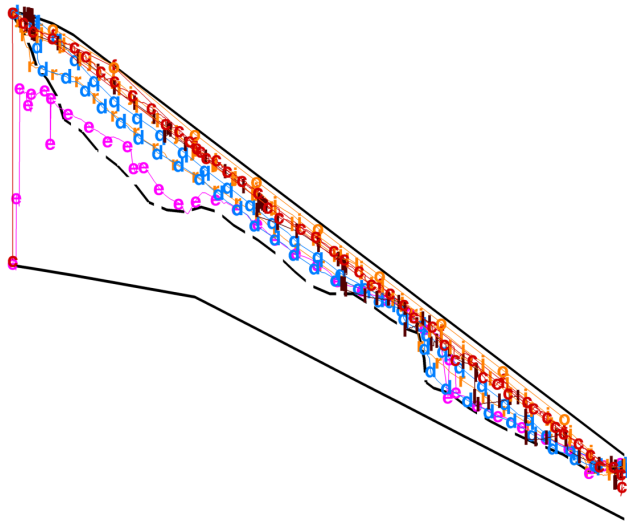
Case 4B – Alpha Sweep



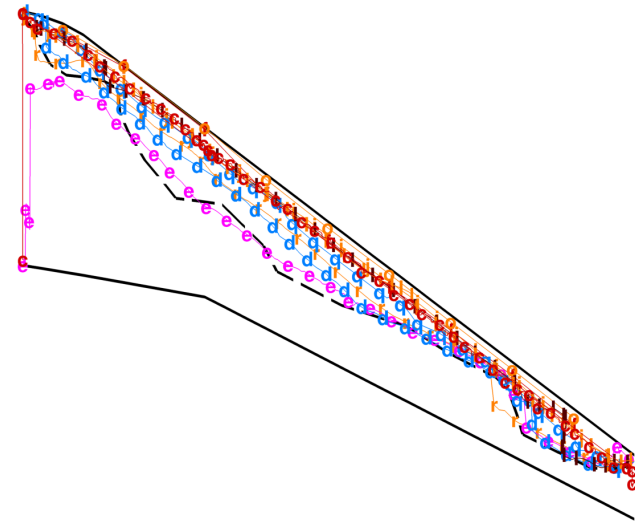
Case 4B – Alpha Sweep



Case 4B – Transition Fronts

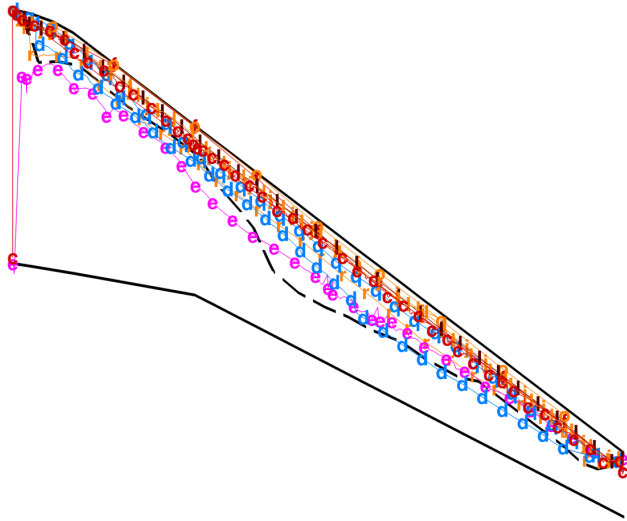


1.45°

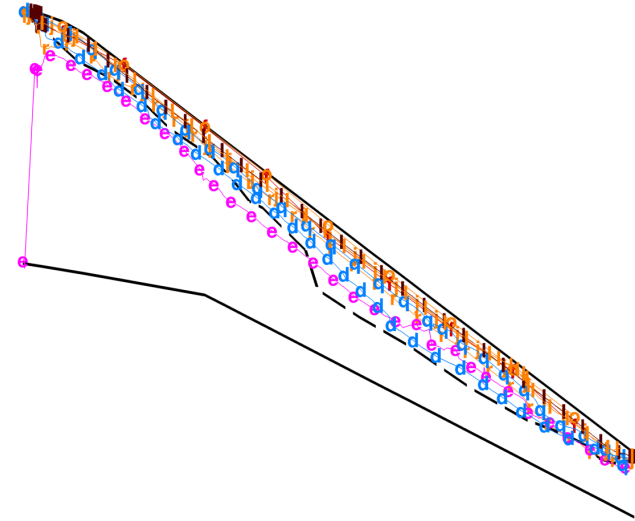


1.98°

Case 4B – Transition Fronts

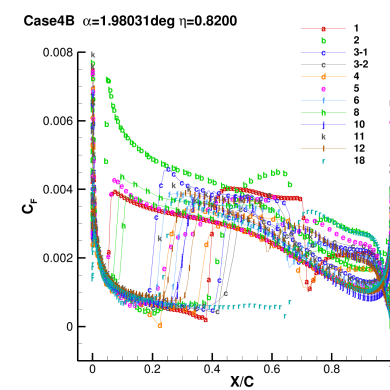
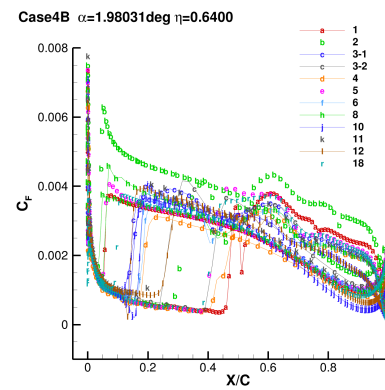
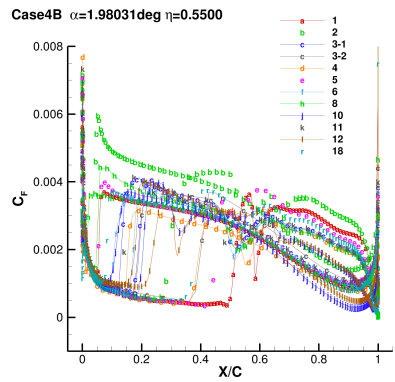
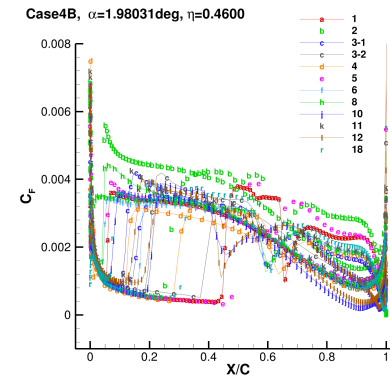
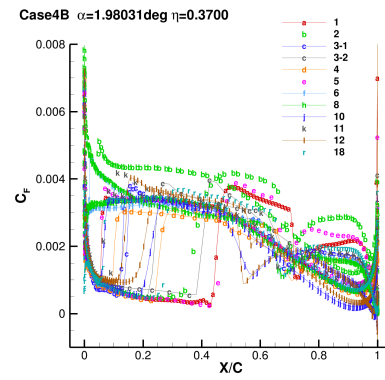
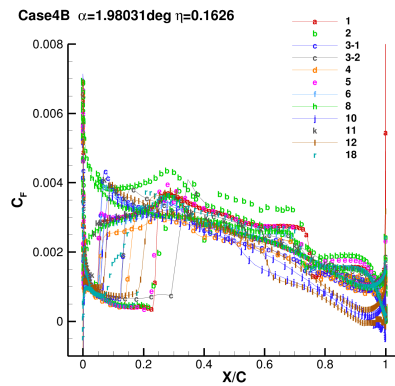


2.46°

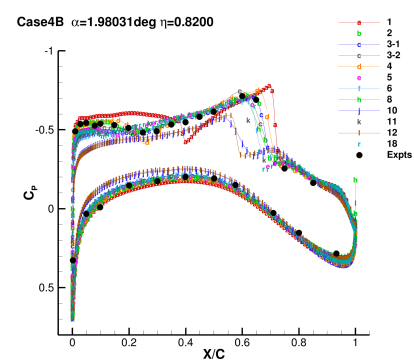
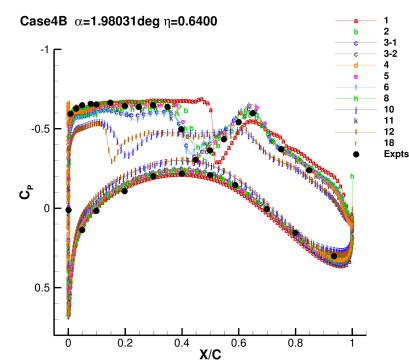
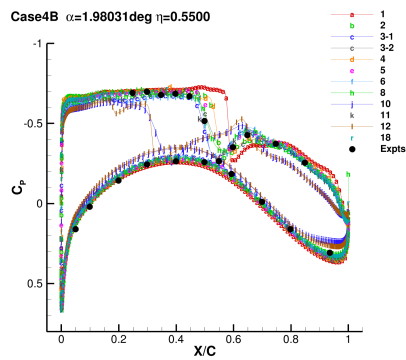
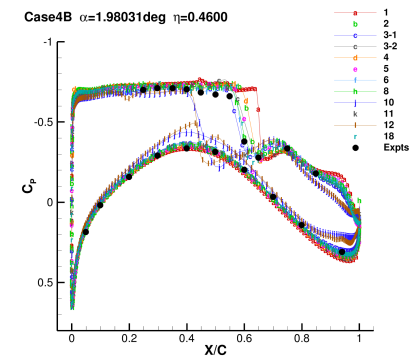
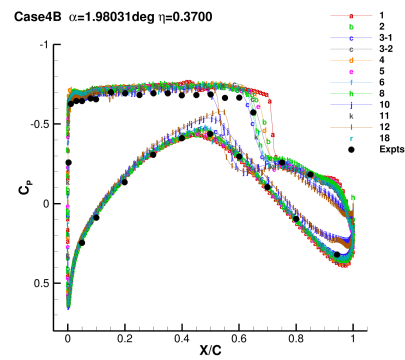
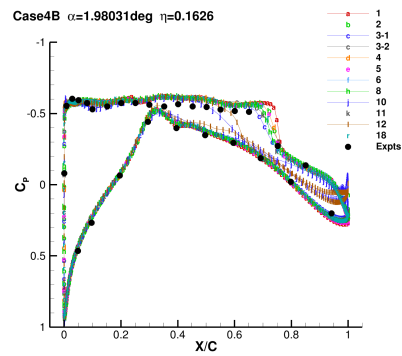


2.94°

Case 4B – Skin-Friction Distributions at 1.98°

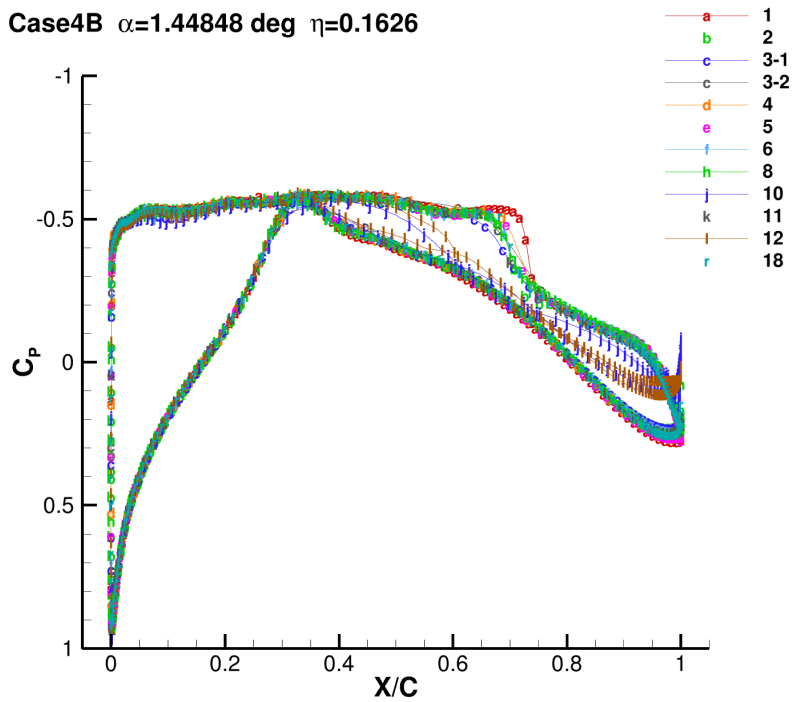


Case 4B – Pressure Distributions at 1.98°

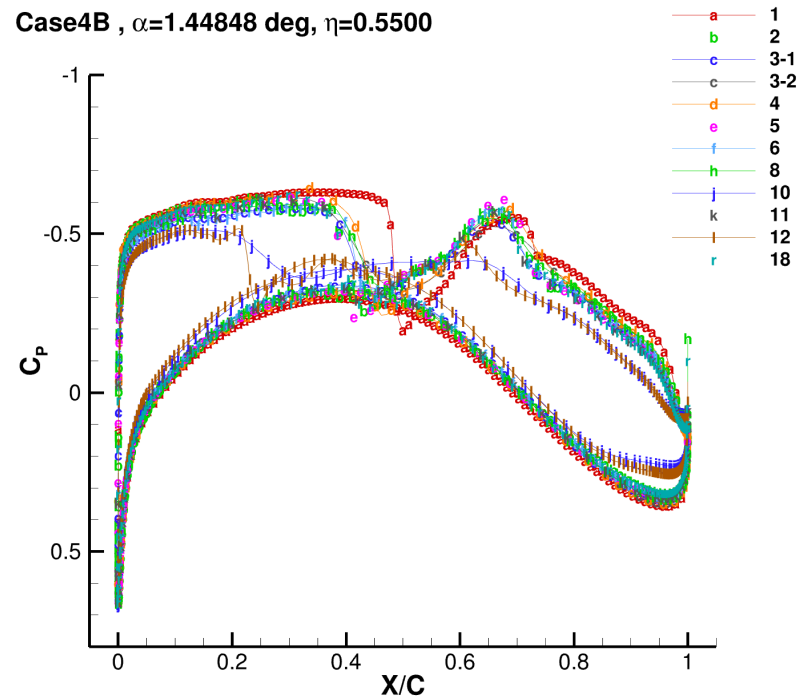


Case 4B – Pressure Distributions

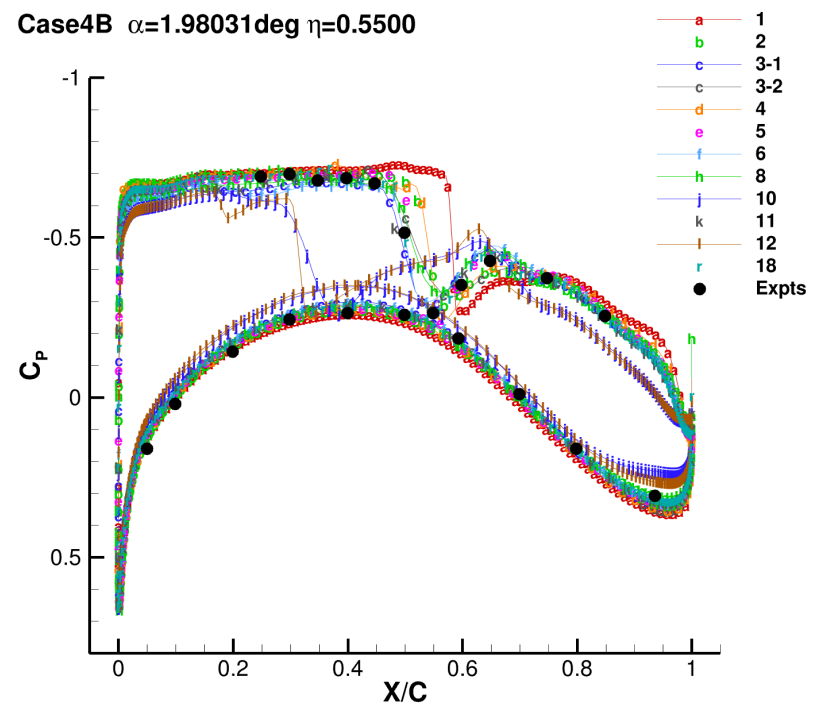
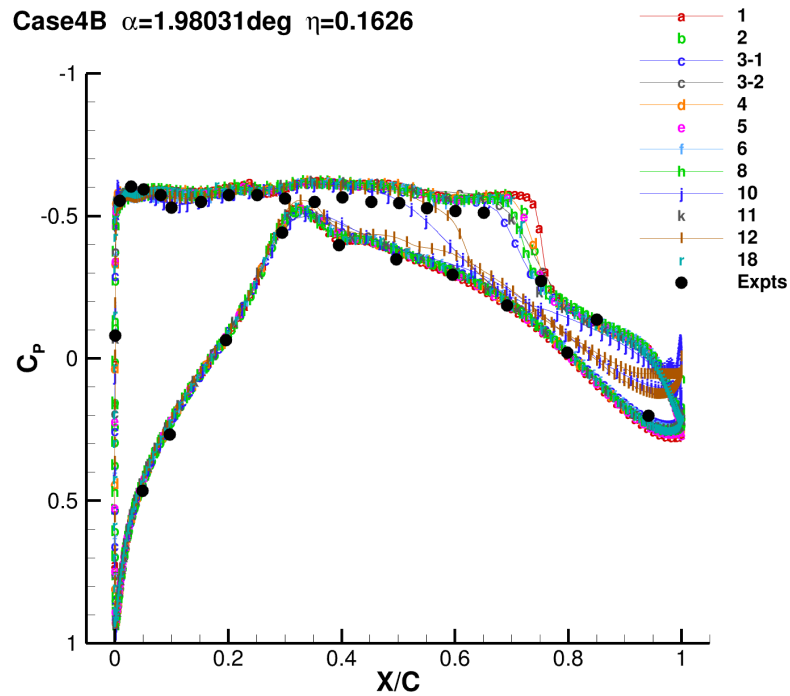
Case4B $\alpha=1.44848$ deg $\eta=0.1626$



Case4B , $\alpha=1.44848$ deg, $\eta=0.5500$

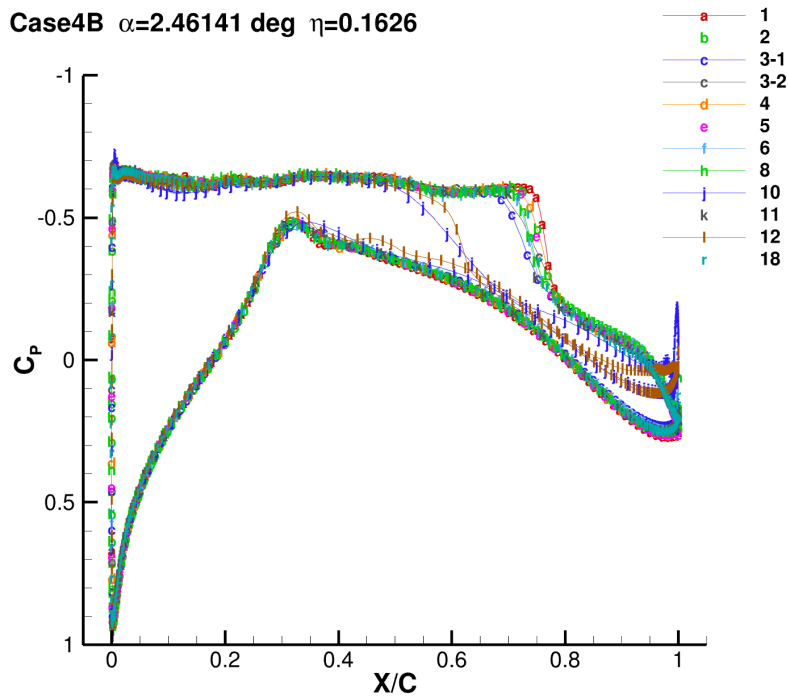


Case 4B – Pressure Distributions

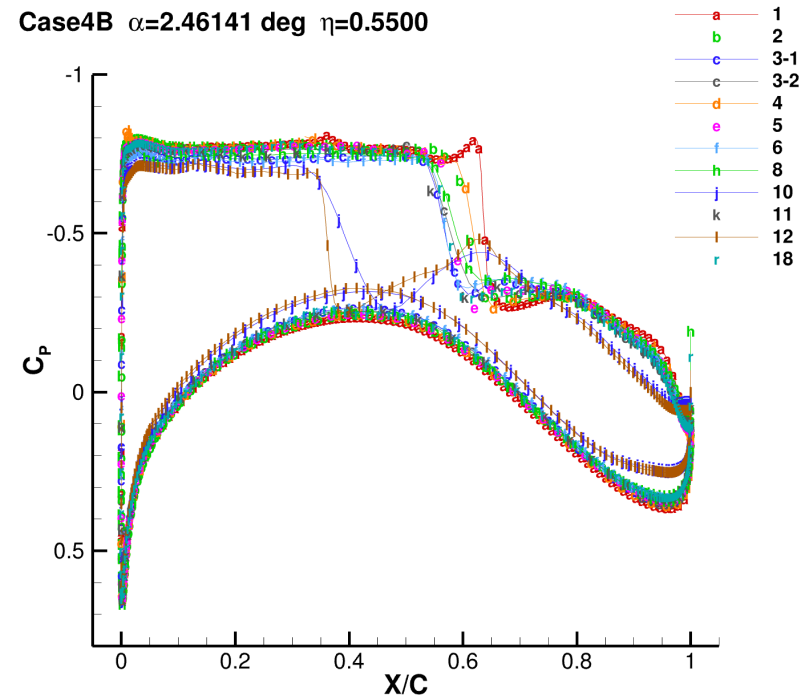


Case 4B – Pressure Distributions

Case4B $\alpha=2.46141$ deg $\eta=0.1626$

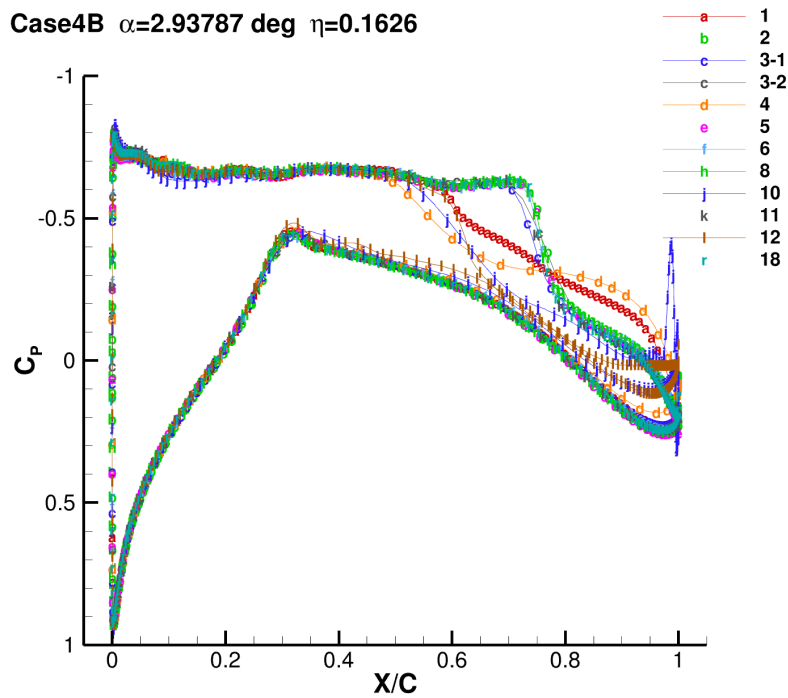


Case4B $\alpha=2.46141$ deg $\eta=0.5500$

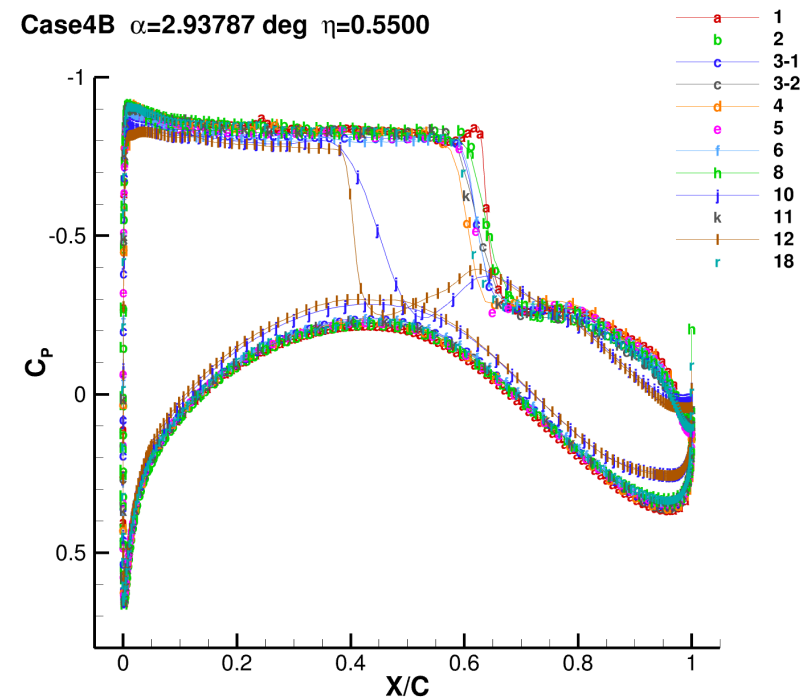


Case 4B – Pressure Distributions

Case4B $\alpha=2.93787$ deg $\eta=0.1626$



Case4B $\alpha=2.93787$ deg $\eta=0.5500$

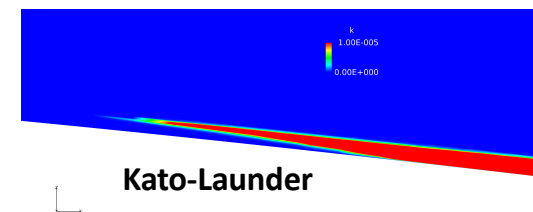
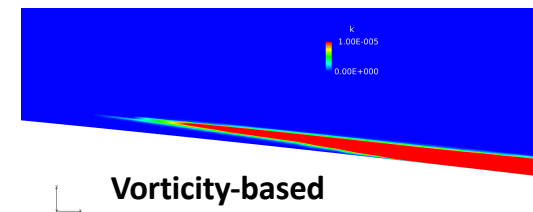
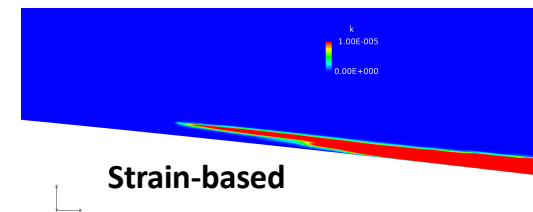


Observations and Next Steps

Code Verification

- **Transition model verification is still an open question**
 - Many custom variants within each modeling framework
 - Coupling with different turbulence models can lead to very different results on the same case
 - Even subtle variants have an impact
 - Strong coupling between transition point and mean flow

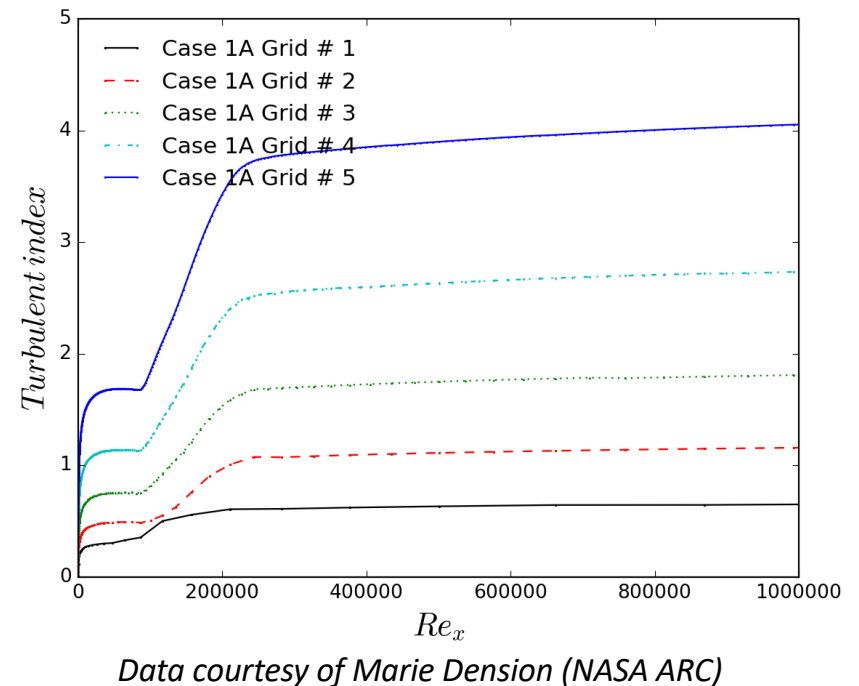
NLF(1)-0416, SST-LM2009



Data courtesy of Jared Carnes (UTK)

Code Verification

- Transition location is not a robust verification metric
 - Too much uncertainty for verification
 - Theoretical shortcomings of turbulence index identified
- Skin-friction at fixed location is better, but...
- **Participants need to have the same models implemented!**



Model Validation

- RANS-based transition modeling is reduced-order by its nature
 - Lack a well-stated PDE like we have for TKE
- Model performance is dependent on the calibration reference
 - Machine Learning provides an opportunity, but needs knowledgeable user
- We have incomplete characterization of test conditions
- **What is the “truth” answer we are aiming for?**

Assessment of Objectives

- ✓ • Assess the current state of the art in laminar-turbulent transition prediction in an industrial CFD environment
- • Determine and document best practices for transitional flow simulations
- ✗ • Verify transition/turbulence model implementations
- ✓ • Encourage risk taking by participants and promote improvements to CFD prediction capabilities

Next Steps

- Participants are encouraged to submit updated results, and presentations will be posted very soon (release pending)
- Workshop-focused Special Session at AIAA SciTech 2022
 - Disseminate workshop outcomes to a broader community
 - Encourage participants to document their results as technical papers
- “Workshop 1.5” targeted for Special Session at Aviation 2022
 - Focus on numerics and model verification

Acknowledgments

- All participants, panelists, attendees, and organizers for helping to bring this to fruition
- Michelle Lynde (NASA LaRC) for assisting with CRM-NLF data and information
- Steve Karman of Pointwise for generating the unstructured CRM-NLF meshes
- AIAA for hosting and being incredibly flexible during the planning

**The 1st AIAA CFD Transition Modeling and Prediction
Workshop was a success**